



NCIE-002

System Architecture & Technical Design

**In development — for human review of surgical
amendment — Version 0.5**

Defines the overall technical architecture of the National Communications Intelligence Ecosystem (NCIE)

Authoritative technical reference for NCIE-003 through NCIE-018 and Codex implementation

Traces to NCIE-001 Master Product Requirements Document, Version 1.3

Prepared: 03 September 2026

Maintained by: National Communications Intelligence Ecosystem (NCIE)

Document Governance

Document	NCIE-002 System & Solution Architecture Specification
Edition	In development — Version 0.5 (VPF & Dynamic Agent Synthesis Architectural Consistency Amendment)
Supersedes	NCIE-002 v0.4 Production Ready (17 August 2026), which itself superseded the Skeletal Architecture Review Edition v0.4 (chapter structure preserved throughout)
Prior baseline	NCIE-002 v0.4 Production Ready (17 August 2026) — historically preserved; see §1.8
Authoritative source	NCIE-001 Master Product Requirements Document, Version 1.3 (VPF & Dynamic Agent Synthesis Consistency Amendment)
Amendment authority	NCIE-007 AI, Agent & Model Orchestration Specification, Version 1.1
Chapter count	32 (revised up from the 20 recorded in the NCIE Production Documentation Suite register; see §1.4). Chapter structure unchanged by this amendment.
Owner	National Communications Intelligence Ecosystem (NCIE) Architecture Function
Assistant identity	Presented to users as ARGUS — Adaptive Regulatory Governance & Unified Surveillance (Ch.19 §19.11)
Reviewer	David King Boison (PhD) — Academic Intelligence Center
Status	IN DEVELOPMENT — FOR HUMAN REVIEW OF SURGICAL AMENDMENT. Not Approved. The v0.4 approval remains historically preserved until v0.5 is reviewed and approved.

1.4 Naming Convention: ARGUS

The NCIE Intelligence Assistant is presented to users under the institutional pseudo-name ARGUS — Adaptive Regulatory Governance & Unified Surveillance. ARGUS is a human-friendly invocation and presentation identity only; it is not a separate intelligence system, and it does not replace “NCIE Intelligence Assistant” as the underlying architectural identity used throughout this specification's component names, interfaces and data model. The full naming boundary, configurability and audit-identity rules governing ARGUS are defined in Chapter 19 §19.11. Wherever this document refers to “the Assistant” or “the NCIE Intelligence Assistant,” ARGUS is that same component under its user-facing name.

1.5 Chapter Count Correction

The NCIE Production Documentation Suite register (17 August 2026) recorded NCIE-002 at 20 chapters. During skeletal review, the reviewing architect identified missing institutional capabilities — most materially active AI collaborative brainstorming (Human ↔ Human ↔ NCIE) rather than invoked-only Assistant access — which required new and expanded chapters. The controlled skeleton was revised to Version 0.4 with 32 chapters. This production-ready edition treats 32 as authoritative; the Suite register chapter count is stale and should be corrected in a subsequent revision of that register.

1.6 How Open Review Items Are Handled

Every chapter of the approved skeleton carried a “Human Review Focus” note flagging institutional decisions that architecture alone cannot settle — hosting model, disaster-recovery targets, acceptance authorities, participation-mode defaults, retention periods, and similar policy calls. Rather than leave these as open questions, this edition resolves every one of them with a specific, reasoned proposed default, marked inline wherever it appears with the following convention:

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. This marker means the preceding statement is an architecturally sound default, not a ratified institutional decision. It must be reviewed and either confirmed or overridden by the applicable NCIE authority before the affected control is treated as binding.

Proposed defaults are chosen to be conservative, auditable and consistent with the mandates already fixed in NCIE-001 (Human-Primary authority, evidence/provenance-first, AI shall not approve NCIE for production, etc.). None of them weaken a control already stated as mandatory in NCIE-001.

1.7 How to Read This Document

Each chapter opens with a scope statement, then proceeds through architecture principles, components and responsibilities, interfaces, data model, event/sequence flows, security controls, failure modes, non-functional

requirements, deployment considerations, acceptance criteria and traceability back to NCIE-001. Diagrams render as structured tables (flow or relationship) rather than free-form graphics so the document remains identical across Word, PDF and any downstream Codex ingestion.

1.8 Document Amendment History

Version: v0.4

Change: Production-ready expansion of the approved 32-chapter skeleton, including the active-AI-collaborative-brainstorming amendment (Ch.19/20/21/23/24).

Status: APPROVED (historical baseline preserved unchanged).

Version: v0.5

Change: VPF & Dynamic Agent Synthesis Architectural Consistency Amendment (Chapters 1, 19, 20, 24, 28, 31, 32).

Authority: Approved NCIE-001 v1.3 and approved NCIE-007 v1.1.

Scope: Controlled surgical amendment — not a rewrite, restructuring or general architecture refresh of NCIE-002. The 32-chapter structure is unchanged.

Architecture Impact: Adds Agent Factory, Agent Registry/Capability Registry, Ephemeral Agent Runtime, VPF, VPF Gate A/B, VPF Profile Registry, and logical Control Plane/Agent Execution Plane boundaries; extends degraded-mode, acceptance and Codex-handover architecture accordingly.

Non-Impact: No change to Human-Primary authority, existing domain architecture, Model Gateway authority, Tool Gateway authority, Evidence governance, Decision authority or Memory ownership.

Status: v0.5 IN DEVELOPMENT — FOR HUMAN REVIEW OF SURGICAL AMENDMENT. The v0.4 approval remains historically preserved until v0.5 is reviewed and approved.

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Architecture Mandate, Scope & Design Principles

Defines the architectural mandate and non-negotiable principles translating NCIE-001 v1.2 into buildable systems.

Full production content. Traces to NCIE-001 v1.2 Product Vision, Mission, Principles & Scope, and Product Scope, Functional Boundaries & NCIE Intelligence Domain Architecture.

1.1 Architectural Mandate

NCIE-002 translates the product commitments fixed in NCIE-001 into a buildable system of record. It is binding on every downstream specification in the suite (NCIE-003 through NCIE-018): where a later document proposes a technology, pattern or interface that conflicts with a principle in this chapter, this chapter governs unless a documented architecture change record supersedes it.

- NCIE is a unified national intelligence environment, not a federation of independent tools presented behind one shell.
- Authority over decisions, findings and enforcement actions is Human-Primary; AI and automation act in advisory, preparatory or executing-under-authorization capacities only.
- Every domain adapter and provider integration is replaceable without redesigning the platform (provider neutrality).
- Every piece of intelligence exposed to a user is traceable to its originating source, rule version and transformation history (evidence/provenance-first).
- Boundaries between raw source data, validated data, canonical intelligence and AI-generated reasoning are explicit and never silently blurred.

1.2 Scope Model

NCIE-002 covers the full technical stack required to deliver the NCIE-001 product scope: the application shell and Situational Awareness workspace; identity and authorization; the canonical data platform; domain intelligence adapters for Network, Topology/GIS, QoS Campaigns, Incidents, Traffic, Revenue, Mobile Money, SIM Registration and Anti-Fraud; regulatory case workflow; cross-domain fusion; alerting; the Intelligence Assistant and collaborative brainstorming; automation/RPA orchestration; rules and calculation governance; evidence and audit; the AI platform (ARGUS, Model Gateway, Tool Gateway, Context/Memory Services, Evaluation/Guardrail Services, and — per the v0.5 amendment — the Agent Factory, Ephemeral Agent Runtime, VPF and Agent/VPF Profile Registries, Ch.24); integration contracts; security and sovereignty; observability; resilience and DR; deployment and DevSecOps; non-functional architecture; testing and acceptance; and operations/Codex handover.

Out of scope for NCIE-002: subscriber-level Mobile Money data (excluded by NCIE-001 product scope), any capability not stated or reasonably implied by NCIE-001, and organizational/staffing decisions, which belong to NCIE-018 handover governance rather than technical architecture.

1.3 Design Principles

1.3.1 Human-Primary Authority

Every architectural component that produces a finding, decision, enforcement action or irreversible state change routes through an explicit human authorization point. AI and RPA components are designed so that removing model or bot availability degrades convenience, not institutional authority.

1.3.2 Provider Neutrality

Network management systems, RPA providers, AI model providers, mapping providers and messaging providers are integrated behind capability-abstracting adapter interfaces (see Chapter 6, Chapter 21, Chapter 24). No chapter of this specification names a specific commercial provider as architecturally load-bearing; provider selections are configuration, not design.

1.3.3 Evidence and Provenance First

Every canonical fact carries source, acquisition time, validation status and applicable rule version. This is a platform-level invariant enforced by the Canonical Data Platform (Chapter 5) and the Evidence, Provenance, Audit & Decision-Time Reconstruction architecture (Chapter 23), not an optional per-domain feature.

1.3.4 Clear Source-to-Decision Boundaries

Five distinct states are never merged into one: raw acquired data; validated/canonical data; AI-generated reasoning or hypotheses; human-adopted working analysis; and formally validated Findings/Decisions. Promotion from one state to the next is an explicit, audited action, never an implicit side effect of storage or display.

1.3.5 Graceful Degradation

Loss of the AI platform, an RPA provider, the mapping provider, or any single source system degrades the affected capability only. Core NCIE — authentication, canonical data access, evidence retrieval, workflow state — remains available. This principle is detailed per-domain in Chapter 28.

1.3.6 AI/Agent Governing Invariants (v0.5 — VPF & Dynamic Agent Synthesis)

Dynamic Agent Synthesis and the VPF validation architecture (Ch.24 §24.4A) extend the AI platform recognized in §1.2; they do not replace the Model Gateway, Tool Gateway, current-authorization or Human-Primary controls already fixed above. The following invariants bind every chapter that touches AI orchestration (principally Ch.19, Ch.20, Ch.24, Ch.28, Ch.31, Ch.32) and are stated once here rather than re-derived per chapter:

- AGENT CREATION ≠ AUTHORITY CREATION.
- AGENT SYNTHESIS ≠ PERMISSION EXPANSION.
- EPHEMERAL AGENT ≠ REGISTERED REUSABLE AGENT.
- AGENT SYNTHESIS ≠ SELF-REGISTRATION.
- SYNTHESIZED AGENT ≠ AGENT-FACTORY AUTHORITY.
- VPF VALIDATION ≠ HUMAN / INSTITUTIONAL APPROVAL.
- UNVALIDATED SUBSTANTIVE AI OUTPUT ≠ VALIDATED NCIE AI RESULT.

1.4 Non-Functional Drivers

Table 1. Non-functional drivers and their binding architectural consequences.

Driver	Architectural Consequence
Auditability	Every write to canonical state carries actor identity (human/AI/bot/service), timestamp and source reference (Ch.23).
Multi-tenancy of trust	Aggregate views and individual/protected views are architecturally distinct data paths, not a display-layer filter (Ch.4, Ch.14).
National-scale data volume	Canonical stores and analytical layers are horizontally scalable and partitioned by operator/geography/time (Ch.5, Ch.30).
Operational resilience	No single external dependency (AI, RPA, map, one source) is a hard availability dependency for core NCIE (Ch.28).
Regulatory defensibility	Decision-time reconstruction must be possible without relying on live external services (Ch.23).

1.5 Architecture Overview

Figure 1. NCIE end-to-end architectural layering, source to human authority.

01	Source Systems & Providers (NMS, Billing, Traffic, Mobile Money, SIM Registration, Anti-Fraud, Email, Maps, Regulatory)
02	Acquisition & Ingestion Layer (Source Registry, API/File/RPA adapters, Ingestion Gateway)
03	Canonical Data Platform (raw / validated / canonical / analytical layers, entity registry, rules)

04	Domain Intelligence Services (Network, Topology, Traffic, Revenue, Mobile Money, SIM, Anti-Fraud, Regulatory)
05	Fusion, Evidence & Alerting (Cross-Domain Fusion, Evidence/Provenance, REWS/Situations)
06	Experience Layer (Situational Awareness Workspace, Intelligence Assistant, Collaborative Rooms)
07	Human Authority (Analysts, Investigators, Regulators, Executives, Auditors)

1.6 Security, Privacy, Sovereignty & Resilience Posture

This chapter states posture only; controls are specified in Chapter 4 (identity/authorization), Chapter 26 (security/privacy/sovereignty), Chapter 28 (resilience/DR) and Chapter 31 (testing/acceptance). Architecturally, NCIE is designed as a sovereign, on-premises-capable national intelligence system: no chapter of this specification requires data to leave Ghanaian jurisdiction as a precondition of core operation.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Default hosting posture is government/NCA-controlled data centre or sovereign cloud within Ghana, with any external AI processing subject to the data-minimisation and residency controls in Chapter 24 and Chapter 26. Confirm the specific hosting facility and sovereign-cloud provider (if any) with NCA IT leadership; this does not change the architecture, only its deployment target (Ch.29).

1.7 Requirements Traceability Approach

NCIE-001 requirements are identified by chapter.paragraph number (e.g. §18.241), which is the stable requirement ID used throughout the suite. Each NCIE-002 chapter closes with a Traceability note mapping its principal components back to the NCIE-001 chapters that mandate them. A consolidated Requirements Traceability Matrix is produced in Chapter 31.

Traceability: NCIE-001 §Product Vision, Mission, Principles & Scope; §Product Scope, Functional Boundaries & NCIE Intelligence Domain Architecture; NCIE-001 v1.3 §12.77-§12.80A, §18.241A; NCIE-007 v1.1 front matter §0.3.

System Context, Actors, Trust Boundaries & External Systems

Maps every user, external system, provider and trust boundary around NCIE.

Full production content. Traces to NCIE-001 v1.2 Product Scope & Functional Boundaries; Security, Privacy, Identity, Authorization & Intelligence Sovereignty Governance.

2.1 Actor Catalogue

Table 2. NCIE actor classes and their trust characteristics.

Actor Class	Representative Roles	Trust Characteristics
Executive	Director-General, Deputy Directors, Divisional Heads	Broad read access to aggregate and Situational Awareness views; no direct data-entry authority.
Analyst	Network, Revenue, Fraud, Regulatory analysts	Domain-scoped read/write on working analysis; cannot validate own Findings.
Engineer	NMS/topology engineers, platform engineers	Elevated access to technical/operational data; no regulatory decision authority.
Regulator / Case Officer	Regulatory Intelligence, compliance officers	Case and Finding/Decision workflow authority within assigned domains and separation-of-duties limits.
Investigator	Anti-Fraud/SIMBOX investigators	Access to sensitive joins and protected-identity resolution under heightened audit (Ch.14, Ch.15, Ch.26).
Administrator	Platform, identity, rule administrators	Configuration and user/rule administration authority; no analytical decision authority by virtue of the admin role alone.
Auditor	Internal/external audit, oversight bodies	Read-only access to evidence, provenance and audit trails; no operational write access.
Machine Identity	RPA bots, scheduled jobs, service accounts, the AI Gateway	Distinct identity class from humans; scoped, time-bounded credentials; never inherits a human's session (Ch.4).

2.2 External System Catalogue

Table 3. External systems and the nature of their exchange with NCIE.

External System	Direction	Nature of Exchange
Operator NMS/PM platforms	Inbound	Performance-monitoring files/exports per operator, per NCIE-001 NMS/PM requirements.
Billing systems	Inbound	Reported Revenue/Billing figures for verification against calculated Revenue.
Traffic data sources	Inbound	International, Off-Net and On-Net traffic volumes by operator/period.
Mobile Money aggregate feeds	Inbound	Daily aggregate sent/received figures by operator; no subscriber-level data.
SIM Registration sources	Inbound	Registration counts and identity-document class distributions.
Anti-Fraud/SIMBOX detection feeds	Inbound	Detection and suspicion signals feeding investigation workflow.
Operator incident mailboxes	Inbound	Outage/fibre-cut/restoration notifications, structured by the Incident Intelligence adapter.

External System	Direction	Nature of Exchange
Mapping/GIS provider	Bidirectional	Base map tiles and geocoding context; NCIE remains the authority for topology and cell placement.
Regulatory Intelligence platform	Bidirectional	Existing case/finding workflow reused rather than rebuilt (Ch.16).
RPA providers (UiPath, Power Automate, Automation Anywhere, future)	Bidirectional	Automated acquisition/execution behind the Capability Registry (Ch.21).
AI model providers (internal/external)	Bidirectional	Conversational and reasoning support behind the Model Gateway (Ch.24); never a system of record.

2.3 Trust Boundaries

Five trust boundaries recur throughout this specification and are referenced by number in later chapters:

- TB-1 — Public/operator network to NCIE ingestion edge (Source Registry, Ingestion Gateway).
- TB-2 — Ingestion edge to raw store (unauthenticated file/API content is never trusted as canonical until validated).
- TB-3 — Human/AI/bot identity boundary at the authorization layer (Ch.4); AI and bots never inherit human session privilege.
- TB-4 — Aggregate-view boundary vs protected-identity boundary (Ch.4, Ch.14); aggregate access does not imply individual-record access.
- TB-5 — NCIE platform boundary to external AI/RPA providers; data crossing this boundary is minimised and governed (Ch.24, Ch.26).

2.4 System Context Diagram

Figure 2. Primary NCIE system-context relationships and the trust boundaries they cross.

SOURCE	RELATIONSHIP	TARGET
Operators (NMS/Billing/Traffic/MoMo/SIM/Fraud)	sends data across TB-1 to	Source Registry / Ingestion Gateway
Operator Incident Mailboxes	feeds across TB-1 to	Incident Intelligence Adapter
Mapping/GIS Provider	supplies base layers across TB-5 to	Topology & Spatial Service
RPA Providers	execute Runs across TB-5 for	Automation Orchestrator
AI Model Providers	serve requests across TB-5 to	Model Gateway
Human Actors	authenticate across TB-3 into	Identity & Authorization Service
Regulatory Intelligence Platform	exchanges cases/findings across TB-5 with	Regulatory Case Service

2.5 Interfaces

Inbound interfaces to source systems are governed by the Source Registry (Ch.6). Outbound interfaces to humans are governed by the Situational Awareness Workspace (Ch.3) and the Intelligence Assistant (Ch.19). All inter-service interfaces internal to NCIE are governed API/event contracts (Ch.25).

2.6 Security Controls

- Every external system is registered in the Source Registry or Capability Registry with a named owner before it can write to NCIE.
- No external system is granted write access to canonical or analytical stores directly; all writes pass through validation (Ch.5, Ch.6).
- Machine identities crossing TB-5 use short-lived, scoped credentials distinct from any human's identity (Ch.4).

2.7 Failure Modes

- Unreachable or malformed source feed: ingestion marks source health degraded (Ch.6, Ch.27); downstream views show missing rather than zero.
- Compromised or decommissioned external system credential: revoked at the Capability Registry/Identity Service without requiring a platform redeploy.

2.8 Acceptance Criteria

- Every actor class in §2.1 maps to at least one authorization role tested in Chapter 31.
- Every external system in §2.2 has a registered owner, expected frequency and health status visible in the Observability platform (Ch.27).

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. The specific list of NCA organisational units mapped to each actor class (§2.1) should be confirmed by NCA HR/organisational leadership; the actor classes themselves are fixed by NCIE-001 and do not change.

Traceability: NCIE-001 §Product Scope, Functional Boundaries & NCIE Intelligence Domain Architecture; §Security, Privacy, Identity, Authorization & Intelligence Sovereignty Governance.

Full-Screen UX & Situational Awareness Workspace

Defines the main operating experience.

Full production content. Traces to NCIE-001 v1.2 NCIE Command Center; Situational Awareness, Early Warning, Alerting & National Communications Operating Picture.

3.1 Experience Principles

- The workspace is full-screen by default; the operator should never need to leave NCIE to check a second system for authoritative state.
- Map, ARGUS (the NCIE Intelligence Assistant), alerts, evidence and domain panels share one operator/geography/time context — changing context in one panel updates the others.
- Views are role-aware: an Executive's default composition differs from an Analyst's or Investigator's, but the underlying data contract is identical.
- Status is always visible in plain language: known, missing, stale, waiting, or contradictory — never a bare blank space.

3.2 Principal Components

Table 4. Situational Awareness Workspace principal components.

Component	Responsibility
Application Shell	Hosts panel composition, global navigation, session and context state; framework-agnostic contract with domain modules.
Workspace Manager	Persists and restores per-user/per-role panel layouts; enforces which panels a role may add.
Dashboard Composition Service	Assembles the National Operating Picture from REWS/Situation, domain health and KPI summaries (Ch.18).
Map Context Bridge	Synchronises map viewport/selection with ARGUS's conversational context and open panels (Ch.8, Ch.19).
Panel Registry	Declares available panels (Network, Topology, Traffic, Revenue, MoMo, SIM, Anti-Fraud, Regulatory, Evidence, Alerts) and their required permissions.

3.3 Context Propagation Flow

Figure 3. Cross-panel context propagation in the Situational Awareness Workspace.

01	Operator selects geography/operator/date in any panel
02	Workspace Manager publishes context change
03	Map Context Bridge updates map viewport and selection
04	Assistant Context Resolver updates conversational grounding (Ch.19)
05	Subscribed domain panels re-query canonical/analytical data for new context
06	National Operating Picture status badges refresh

3.4 Data Model

- Workspace Layout — user/role, panel set, positions, saved filters.

- Context State — active operator(s), geography, date/time range, selected entity references (transient, not persisted as evidence).
- Panel Contract — panel id, required permission, supported context dimensions, data source binding.

3.5 Security Controls

- Panel visibility is authorization-filtered server-side, not merely hidden client-side (Ch.4).
- Context state never carries protected-identity fields (Ch.14) into panels the active user is not authorized to view.

3.6 Failure Modes

- Map provider unavailable: workspace falls back to a non-map, list/table-based geography selector; all non-map panels remain fully functional (Ch.28).
- One domain panel's data source is degraded: only that panel shows a degraded-status badge; other panels are unaffected.

3.7 Non-Functional Requirements

- Initial workspace load renders the shell and cached last-known status within 2 seconds on the reference network profile (Ch.30 defines the profile).
- Context propagation across panels completes within 500ms of a context change under normal load.

3.8 Acceptance Criteria

- For each role class in Ch.2, a default panel composition exists and is demonstrably restricted to authorized panels only.
- Changing operator/geography/date in any one panel is shown to update all subscribed panels within the NFR target.

3.9 ARGUS Presentation

ARGUS — the user-facing name for the NCIE Intelligence Assistant (Ch.19 §19.11) — is presented within the workspace, not as a separate application. Supported presentation states are:

- Docked — a persistent side panel alongside other workspace panels.
- Expanded — a larger conversational/collaborative pane while other panels remain visible.
- Full-Screen — ARGUS occupies the primary workspace area.
- Present inside Collaborative Rooms — participating per its current Participation Mode (Ch.20 §20.7).

3.10 UX Doctrine

Figure 4. The governing NCIE UX doctrine loop, from national situational awareness to human-governed action.

01	See the national situation
02	Understand what is current, stale, missing or conflicting
03	Select the operator / geography / object
04	Drill into evidence
05	Ask ARGUS
06	Brainstorm with humans + ARGUS
07	Identify hypotheses / contradictions / evidence gaps
08	Launch or monitor governed evidence acquisition
09	Receive new validated intelligence

	▼
10	Update the map / situation / reasoning
	▼
11	Move into human-governed action where required

Colour never substitutes for meaning; frontend state never substitutes for authoritative state; an ARGUS suggestion never displays as a completed action; and a hypothesis never displays as a Finding.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Default panel compositions for Executive, Analyst, Engineer, Regulator/Case Officer, Investigator, Administrator and Auditor roles are proposed in NCIE-012 (UX/UI & Design System Specification); confirm final layouts with representative users from each role before production rollout.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Default ARGUS presentation state (§3.9) is proposed as Docked for ordinary workspace use, Expanded when a collaborative room becomes the active panel, and Full-Screen only on explicit user action; confirm with representative users before production rollout.

Traceability: NCIE-001 §NCIE Command Center; §Situational Awareness, Early Warning, Alerting & National Communications Operating Picture.

Identity, Authorization & Zero-Trust

Defines authentication and permission enforcement.

Full production content. Traces to NCIE-001 v1.2 Security, Privacy, Identity, Authorization & Intelligence Sovereignty Governance.

4.1 Principles

- Zero-trust: no request is implicitly trusted by network location; every request is authenticated and authorized at the service boundary.
- RBAC is the baseline, extended with contextual domain/object/field/action controls where NCIE-001 requires finer granularity than role alone provides (e.g. protected identity fields).
- Aggregate access never implies individual access; a role permitted to see aggregate MoMo volumes is not thereby permitted to see any subscriber-level data (which is out of scope entirely, Ch.13).
- Collaborative rooms (Ch.20) never pool participant permissions; a room's shared context is filtered per-viewer by that viewer's own authorization.
- Human, AI and machine identities are architecturally distinct identity classes; an AI or bot identity never silently assumes a human's session.

4.2 Principal Components

Table 5. Identity, Authorization & Zero-Trust principal components.

Component	Responsibility
Identity Service	Authenticates humans (SSO/MFA-ready) and issues session tokens; manages machine-identity credentials separately.
Authorization Service	Evaluates RBAC plus contextual domain/object/field/action policy for every request; single source of authorization truth.
Session Manager	Issues, refreshes and revokes sessions; enforces idle/absolute timeout policy.
Machine Identity Registry	Issues scoped, time-bounded credentials to RPA bots, scheduled jobs and the AI Gateway; distinct from human accounts.
Policy Administration Point	Where administrators define/version roles, field-level rules and separation-of-duties constraints (Ch.32).

4.3 Authorization Model

Authorization decisions combine four layers, evaluated in order: (1) role-based domain access, (2) object-level access (e.g. a specific case or room), (3) field-level access (e.g. protected identifiers in Ch.14), (4) action-level access (e.g. may-validate-Finding vs may-propose-Finding). A request is denied if any layer denies it; layers do not override one another upward.

Figure 5. Identity and authorization relationships.

SOURCE	RELATIONSHIP	TARGET
Human Actor	authenticates via	Identity Service
Identity Service	issues session to	Session Manager
Authorization Service	evaluates request against	Policy Administration Point rules
Machine Identity Registry	issues scoped credential to	RPA Bot / AI Gateway / Scheduled Job

4.4 Data Model

- Identity — human/machine, credential reference, MFA status, lifecycle status.
- Role — named bundle of domain/object/field/action grants, versioned.
- Policy Grant — role-to-permission mapping, effective-dated (aligns with Ch.22 temporal governance).
- Session — identity reference, issued/expiry time, device/context metadata, revocation status.

4.5 Security Controls

- SSO/MFA-ready integration; MFA is mandatory for Administrator, Investigator and regulatory validation actions.
- Separation of duties: the role that proposes a Finding cannot also validate the same Finding (Ch.16).
- Break-glass access (Ch.26) is logged, time-bounded and requires post-hoc justification.
- All authorization decisions — permit and deny — are logged for audit (Ch.23).
- Authorization and audit for the NCIE Intelligence Assistant bind to its stable internal service identity, independent of its user-facing presentation name (ARGUS); a future institutional rename does not require reissuing authorization policy or audit history (Ch.19 §19.11).

4.6 Failure Modes

- Identity Service degraded: existing valid sessions continue to function per policy; new sign-ins queue or fail closed, never fail open.
- Policy Administration Point unreachable: Authorization Service uses last-known-good cached policy with a visible staleness indicator; it does not default to permissive.

4.7 Non-Functional Requirements

- Authorization decisions complete within 100ms at p95 under normal load (Ch.30).
- Session revocation propagates to all active service boundaries within 30 seconds.

4.8 Acceptance Criteria

- Attempting an action denied by any one of the four authorization layers is demonstrably blocked, with an audit entry recorded.
- A room with mixed-permission participants is shown to filter shared context per-viewer, not per-room (Ch.20).

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Default session timeouts: 30 minutes idle / 12 hours absolute for standard roles; 15 minutes idle / 8 hours absolute for Investigator and Administrator roles. Confirm against NCA information-security policy before production.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Default MFA method: authenticator-app TOTP as baseline, with SMS fallback disabled by default given SIM-based fraud relevance to NCIE's own domain. Confirm with NCA IT security.

Traceability: NCIE-001 §Security, Privacy, Identity, Authorization & Intelligence Sovereignty Governance.

Canonical Data Platform, Domain Model & Metadata

Defines the shared data foundation.

Full production content. Traces to NCIE-001 v1.2 Data, Intelligence Domain, Canonical Model & Information Governance.

5.1 Layered Data Architecture

- Raw layer — unmodified acquired content (files, API payloads, email bodies, screenshots), immutable once stored.
- Validated layer — raw content checked against source-registry expectations (operator/date/scope correctness); rejected content is flagged, not silently dropped.
- Canonical layer — the single authoritative representation of each entity (operator, geography, cell, SIM, identity, incident, case, evidence), reconciled across sources.
- Analytical layer — derived aggregates, KPIs and fusion outputs optimised for query, always traceable back to canonical sources.

5.2 Canonical Entities

Table 6. Canonical entity registry (summary; full data dictionary in NCIE-004/NCIE-014).

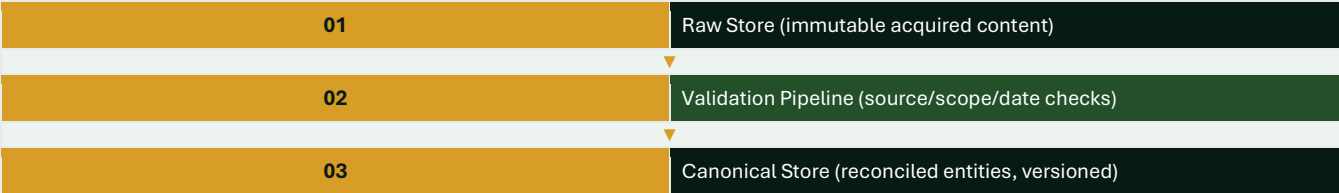
Entity	Description	Owning Domain Chapter
Operator	Licensed telecom operator; stable identifier used across all domains.	Ch.6
Geography	National/regional/district/place hierarchy with versioned boundaries.	Ch.8
Cell	Network cell/site, versioned by topology change.	Ch.7, Ch.8
SIM / Identity	Protected SIM and identity-document reference (Ch.14); never exposed in aggregate views.	Ch.14
Incident	Outage/fibre-cut/restoration event with lineage to source email/NMS evidence.	Ch.10
Case	Regulatory workflow container linking exceptions, findings and decisions.	Ch.16
Evidence	Any authoritative artifact (file, calculation, screenshot, extraction) with provenance.	Ch.23

5.3 Metadata & Temporal Model

- Every canonical record distinguishes event time (when the fact occurred), acquisition time (when NCIE received it) and knowledge time (when NCIE considered it known) — late-arriving data preserves its true knowledge time rather than back-dating silently.
- Corrections are versioned, never overwritten in place; the prior version remains queryable for decision-time reconstruction (Ch.23).
- Missing data is represented as an explicit missing state, distinct from a zero value, across every domain adapter.

5.4 Data Flow

Figure 6. Raw-to-analytical data flow through the Canonical Data Platform.



	▼
04	Analytical Layer (aggregates, KPIs, fusion inputs)
	▼
05	Domain Services & Situational Awareness Workspace

5.5 Metadata Catalogue

A metadata catalogue records, for every canonical field: source(s) of truth, permitted transformations, classification (public/internal/protected/sensitive), and owning domain chapter. The catalogue is the input to both the Authorization Service field-level rules (Ch.4) and the Data Protection controls (Ch.26).

5.6 Security Controls

- Protected and sensitive fields (Ch.14, Ch.15) are tagged in the metadata catalogue and enforced at query time, not only at display time.
- Raw layer retains original content for chain-of-custody purposes even after canonicalisation (Ch.23).

5.7 Failure Modes

- Validation pipeline rejects a batch: the batch is quarantined with a reason; canonical state is unaffected until corrected and re-validated.
- Conflicting canonical values from two sources: both are retained with provenance; the platform does not silently pick one without a resolution rule (Ch.17, Ch.22).

5.8 Non-Functional Requirements

- Canonical store scales horizontally by operator/geography/time partition to national data volumes (Ch.30 sets specific targets).
- Analytical layer refresh lag from canonical write is bounded and visible per domain (Ch.27).

5.9 Acceptance Criteria

- For each canonical entity in §5.2, a versioned change is demonstrably retrievable at a prior point in time.
- A deliberately malformed source batch is demonstrably quarantined rather than corrupting canonical state.

Traceability: NCIE-001 §Data, Intelligence Domain, Canonical Model & Information Governance.

Source Registry, Acquisition & Ingestion

Defines how sources enter NCIE.

Full production content. Traces to NCIE-001 v1.2 Data Acquisition, Integration, Source Intelligence & Ingestion Governance.

6.1 Source vs Provider

A Source is the institutional origin of data (e.g. "Operator X monthly PM export"); a Provider is the technical channel used to acquire it (API, file drop, email, screenshot, manual entry, RPA-driven portal access). The Source Registry tracks both independently so a provider can change (e.g. manual to RPA) without altering the source's identity or history.

6.2 Principal Components

Table 7. Source Registry, Acquisition & Ingestion principal components.

Component	Responsibility
Source Registry	Catalogue of every source: owner, expected frequency, format, provider(s), API-vs-RPA access mode.
Ingestion Gateway	Single technical entry point for API, file, email and RPA-submitted content.
Raw Store	Immutable storage of acquired content pre-validation.
Validation Pipeline	Checks operator/date/scope correctness; produces Request/Run/artifact/validation lineage.
Source Health Monitor	Tracks freshness, last successful acquisition and error state per source (feeds Ch.27).

6.3 Acquisition Lineage

Every acquisition is modelled as a Request (what was asked for), a Run (an attempt to fulfil it), an Artifact (what was actually received), and a Validation outcome. This lineage is preserved even when acquisition fails, so a missing PM file is visible as a failed Run rather than silent absence.

Figure 7. Request/Run/Artifact/Validation acquisition lineage.

SOURCE	RELATIONSHIP	TARGET
Request	produces	Run
Run	yields	Artifact
Artifact	passes through	Validation Pipeline
Validation Pipeline	writes to	Canonical Store

6.4 Validation Rules

- Wrong operator, wrong date range or out-of-scope content is rejected at validation, not silently accepted and miscategorised.
- Missing data remains missing — validation never substitutes a zero or default value for an absent measurement.
- Late-arriving data is accepted and back-referenced to its true event time while preserving true knowledge time (Ch.5).

6.5 Provider Channels Supported

- API — direct programmatic acquisition, preferred where available.
- File — scheduled or manual file drop (e.g. NMS/PM exports).
- Email — structured extraction from operator incident mailboxes (Ch.10).
- Screenshot — image capture with OCR/manual transcription support and explicit provenance flag.

- Manual entry — human-entered data, flagged distinctly from system-acquired data.
- RPA — provider-neutral robotic acquisition via the Automation Orchestrator (Ch.21) for portals without an API.

6.6 Security Controls

- Every source has a named accountable owner in the registry before it is enabled.
- Ingestion Gateway authenticates and rate-limits every channel; unregistered sources are rejected.

6.7 Failure Modes

- Source unreachable: Source Health Monitor marks it degraded; consumers see explicit staleness, not silent gaps (Ch.27).
- Malformed artifact: quarantined with a reason, visible to the source owner for correction.

6.8 Non-Functional Requirements

- Validation pipeline processes standard daily source volumes within the freight window required by downstream KPI/REWS timing (Ch.7, Ch.18).

6.9 Acceptance Criteria

- Every source in the registry has owner, frequency and provider-channel populated and visible in Observability (Ch.27).
- A deliberately wrong-operator file is demonstrably rejected with a clear reason rather than miscategorised.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Default source health thresholds mirror the NMS/PM convention already fixed in NCIE-001 Chapter 7: Green 66-100% expected files received, Amber under 66%, Grey no files received, applied uniformly across all source types unless a domain chapter states otherwise.

Traceability: NCIE-001 §Data Acquisition, Integration, Source Intelligence & Ingestion Governance.

Network Intelligence, NMS, PM & KPI

Defines national-to-cell network intelligence.

Full production content. Traces to NCIE-001 v1.2 Product Capability Architecture; Data, Intelligence Domain, Canonical Model & Information Governance.

7.1 Scope

Covers acquisition and analysis of Performance Monitoring (PM) data from national to cell level, KPI computation, threshold-based health status, and district-to-cell investigation triggered by degradation.

7.2 Principal Components

Table 8. Network Intelligence principal components.

Component	Responsibility
NMS Adapters	Provider-neutral ingestion of operator PM exports (Ch.6 channels).
PM Service	Normalises PM records to canonical cell/KPI/date structure.
KPI Engine	Computes KPIs against explicit definitions and thresholds; versioned KPI definitions.
Investigation Service	Escalates district-level degradation into cell-level investigation workflow.

7.3 KPI Health Status Model

Each geography/cell/date/KPI combination carries an explicit health status derived from expected-vs-received file coverage: Green (66-100% of expected files received), Amber (under 66%), Grey (no files received). This status is distinct from the KPI value itself — a cell can have a good KPI value with Amber coverage, and both are shown.

7.4 District-to-Cell Investigation Flow

Figure 8. District-degradation to cell-investigation workflow.

01	Daily district KPI extraction
02	Degradation detected against threshold
03	Topology resolution (district to constituent cells, Ch.8)
04	Cell-level PM retrieval and KPI recomputation
05	Investigation Service opens investigation record with evidence links
06	Analyst review and escalation to Alert/Situation if warranted (Ch.18)

7.5 Data Model

- PM Record — cell, date, KPI, value, source, coverage status.
- KPI Definition — name, formula, threshold, effective date range (versioned per Ch.22).
- Investigation — trigger geography, opened date, linked cells, linked evidence, status, outcome.

7.6 Security Controls

- KPI definitions are change-controlled through the Rule Service (Ch.22); no adapter may hardcode a threshold.

7.7 Failure Modes

- NMS export delayed: coverage status shows Grey/Amber rather than a false Green; investigation triggers are suppressed until data arrives or a manual override is recorded.
- PM screenshot ingestion (where no export is available) persists partial state and resumes on next delivery without losing prior progress.

7.8 Non-Functional Requirements

- KPI Engine recomputation for a full day's national PM volume completes within the operational window required for next-morning district review.

7.9 Acceptance Criteria

- A synthetic district degradation is demonstrably escalated to cell-level investigation with correct topology resolution.
- Coverage status (Green/Amber/Grey) is shown to be independent of and alongside the KPI value.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm the authoritative KPI set (e.g. availability, drop-call rate, congestion) and their formulas/thresholds with NCA network-engineering leadership; the Rule Service architecture supports any agreed set without redesign.

Traceability: NCIE-001 §Product Capability Architecture (Network Intelligence); §Data, Intelligence Domain, Canonical Model & Information Governance.

Topology, GIS, Spatial Intelligence & Mapping

Defines place-to-cell resolution and mapping.

Full production content. Traces to NCIE-001 v1.2 Topology, Geography & QoS Campaign Intelligence.

8.1 Scope

Covers versioned network topology, natural-language place-to-cell resolution, historical topology reconstruction, and governed map layer presentation.

8.2 Principal Components

Table 9. Topology, GIS & Spatial Intelligence principal components.

Component	Responsibility
Topology Service	Maintains versioned cell/site topology per operator; historical topology retrievable.
Spatial Database	Stores geography, place and cell geometries for spatial queries.
Geocoder	Resolves natural-language place names to canonical geography, using minimal external context only.
Map API	Serves governed, permission-aware map layers to the workspace (Ch.3).

8.3 Place-to-Cell Resolution Flow

This is the reference "Madina-style" flow named in the approved skeleton: a natural-language place name resolves through canonical geography to topology to cells to PM/NMS data to analysis, and it must remain intact end to end.

Figure 9. Natural-language place to cell-level analysis resolution flow.

01	Natural-language place name (e.g. 'Madina')
02	Geocoder resolves to canonical Geography entity
03	Topology Service resolves Geography to constituent Cells (version-aware)
04	PM/NMS Service retrieves KPI data for resolved cells (Ch.7)
05	Situational Awareness Workspace / Assistant present grounded analysis

8.4 Data Model

- Geography — hierarchical place entity (national/regional/district/place), versioned boundary.
- Cell/Site — network element with location, versioned by topology change history.
- Topology Version — snapshot of cell-to-geography and cell-to-site relationships at a point in time.

8.5 Security Controls

- Map layers are permission-filtered per role before rendering; sensitive layers (e.g. Anti-Fraud locations) are never exposed to unauthorized roles even as a visual overlay.

8.6 Failure Modes

- External mapping provider unavailable: NCIE falls back to a non-map geography selector (Ch.3, Ch.28); topology and cell data remain fully queryable.

- Ambiguous place-name resolution: system presents candidate matches for human disambiguation rather than guessing silently.

8.7 Non-Functional Requirements

- Place-to-cell resolution completes within 2 seconds at p95 for common place names.

8.8 Acceptance Criteria

- The Madina-style flow (§8.3) is demonstrably reproducible end to end for a representative set of place names.
- A historical topology query correctly reflects the topology as it existed at a specified prior date.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm the specific external mapping/geocoding provider and licensing model with NCA procurement; the architecture treats it as a replaceable provider behind the Map API and Geocoder (§8.2).

Traceability: NCIE-001 §Topology, Geography & QoS Campaign Intelligence.

QoS Campaign / Drive-Test Intelligence

Defines driven-test evidence reuse.

Full production content. Traces to NCIE-001 v1.2 Topology, Geography & QoS Campaign Intelligence.

9.1 Scope

Covers reuse of drive-test/QoS campaign data: campaign ingestion, geometry matching to topology, and presentation of the latest applicable campaign per geography alongside full historical access.

9.2 Principal Components

Table 10. QoS Campaign Intelligence principal components.

Component	Responsibility
Campaign Repository	Stores QoS/drive-test campaigns with date, operator, route, technology and measurements.
Geometry Store	Holds route/measurement geometries for spatial matching.
Matcher	Associates campaign geometry with topology/geography for geography-specific latest-applicable resolution.
Analytics Service	Computes campaign-derived KPIs and trend comparisons across campaigns.

9.3 Latest-Applicable Resolution

For any given geography, "latest applicable" campaign data is geography-specific: a newer campaign in one district does not make an older campaign in a different, unsampled district stale by association. Age and coverage are always shown alongside the data so a user can judge relevance.

Figure 10. QoS Campaign ingestion-to-analysis flow.

01	Campaign file/API ingestion (Ch.6)
02	Geometry Store receives route/measurement geometry
03	Matcher associates geometry to topology/geography
04	Campaign Repository indexes by operator/date/geography
05	Analytics Service surfaces latest-applicable + full history on request

9.4 Data Model

- Campaign — operator, date, route, technology, measurement set, source.
- Measurement Point — geolocated sample with KPI values.
- Geography Coverage Index — which geographies a given campaign actually sampled.

9.5 Security Controls

- Campaign data is attributed to source operator/vendor and cannot be edited post-ingestion except through a versioned correction (Ch.5).

9.6 Failure Modes

- Campaign geometry fails to match any known geography: retained and flagged unmatched rather than discarded, pending manual topology reconciliation.

9.7 Non-Functional Requirements

- Latest-applicable lookup for a given geography returns within 2 seconds at p95.

9.8 Acceptance Criteria

- For a geography sampled by multiple campaigns over time, the system correctly identifies the latest-applicable campaign and still allows retrieval of all prior campaigns.
- Age and coverage indicators are shown alongside every campaign-derived value in the workspace.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm required campaign formats, KPI set and retention period with NCA network-engineering leadership; the Campaign Repository schema is format-agnostic at the storage layer.

Traceability: NCIE-001 §Topology, Geography & QoS Campaign Intelligence.

Incident Intelligence & Operator Email

Converts daily outage/fibre-cut/restoration email into governed incidents.

Full production content. Traces to NCIE-001 v1.2 Product Capability Architecture; Automation Orchestration & RPA Data Acquisition.

10.1 Scope

Converts daily operator outage/fibre-cut/restoration email into governed, structured incident records, fused with NMS evidence for restoration verification.

10.2 Principal Components

Table 11. Incident Intelligence & Operator Email principal components.

Component	Responsibility
Mailbox Connector	Reads designated operator incident mailboxes; original email is retained as evidence, never modified.
Extractor	Structures operator-reported incident data (type, location, start time, cause) from email content.
Threading Service	Deduplicates and threads related messages (initial report, updates, restoration) into one Incident.
Incident Workflow	Tracks incident status; checks operator-reported restoration against NMS evidence before closure.

10.3 Email-to-Incident Flow

Figure 11. Operator incident email to governed Incident record.

01	Operator incident email arrives in designated mailbox
02	Mailbox Connector retains original as evidence
03	Extractor structures incident fields
04	Threading Service links to existing Incident or opens a new one
05	Incident Workflow cross-checks restoration claim against NMS data (Ch.7)
06	Incident closed or escalated to REWS/Situation (Ch.18)

10.4 Data Model

- Incident — category, affected geography/cells, start/restoration time (reported and verified), status, linked evidence.
- Email Evidence — original message, sender, timestamp, thread linkage; immutable.

10.5 Security Controls

- Email content cannot issue commands to NCIE; the Extractor only produces proposed structured fields for the Incident Workflow, never direct system actions.
- Original email is retained unmodified as evidence for the life of the incident record (Ch.23).

10.6 Failure Modes

- Restoration claimed by email but not corroborated by NMS: incident remains open pending verification rather than auto-closing.

- Mailbox unreachable: Source Health Monitor (Ch.6) flags degraded status; no incidents are silently missed without visibility.

10.7 Non-Functional Requirements

- New incident emails are structured and threaded within 5 minutes of arrival under normal load.

10.8 Acceptance Criteria

- A multi-message incident thread (report, update, restoration) is demonstrably consolidated into one Incident with full lineage to each source email.
- A restoration claim unsupported by NMS evidence is demonstrably kept open rather than auto-closed.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm designated incident mailboxes, incident category taxonomy and closure rules with NCA network operations; the extraction/threading architecture is taxonomy-agnostic.

Traceability: NCIE-001 §Product Capability Architecture; §Automation Orchestration & RPA Data Acquisition.

Traffic Intelligence

Defines International, Off-Net and On-Net Traffic.

Full production content. Traces to NCIE-001 v1.2 Product Capability Architecture; Regulatory Intelligence, Compliance, Revenue Assurance & Governed Case Progression.

11.1 Scope

Covers International, Off-Net and On-Net traffic intelligence: explicit operator/period/type/volume/unit/source/coverage attribution, versioned corrections, and the feed into Revenue calculation (Ch.12) and cross-domain fusion (Ch.17).

11.2 Principal Components

Table 12. Traffic Intelligence principal components.

Component	Responsibility
Traffic Adapter	Ingests traffic volumes per operator/period/type via Ch.6 channels.
Traffic Store	Canonical, versioned traffic records.
Analytics Service	Trend, comparison and anomaly views over traffic volumes.
Quality Service	Flags missing periods and out-of-pattern values for review, without auto-correcting them.

11.3 Data Model

- Traffic Record — operator, period, traffic type (International/Off-Net/On-Net), volume, unit, source, coverage status.
- Correction — versioned adjustment to a prior Traffic Record, with reason and actor.

11.4 Traffic-to-Revenue Flow

Figure 12. Traffic Intelligence feeding Revenue and Fusion.

01	Traffic Adapter ingests operator traffic volumes
02	Quality Service flags missing/anomalous periods
03	Traffic Store holds canonical, versioned traffic
04	Revenue Calculator consumes Traffic for calculated Revenue (Ch.12)
05	Cross-Domain Fusion consumes Traffic for correlation (Ch.17)

11.5 Security Controls

- Traffic classifications (International/Off-Net/On-Net) are fixed reference data, changeable only through the Rule Service change process (Ch.22).

11.6 Failure Modes

- Missing period: remains explicitly missing in Revenue calculation inputs rather than treated as zero traffic (Ch.12).

11.7 Non-Functional Requirements

- Monthly national traffic ingestion for all operators completes within the operational window required for Revenue Verification reporting (Ch.12).

11.8 Acceptance Criteria

- A traffic correction is demonstrably versioned, with the prior value still retrievable for decision-time reconstruction.

Traceability: NCIE-001 \$Product Capability Architecture; \$Regulatory Intelligence, Compliance, Revenue Assurance & Governed Case Progression.

Revenue Intelligence & Billing Verification

Defines deterministic Revenue calculation and reconciliation.

Full production content. Traces to NCIE-001 v1.2 Regulatory Intelligence, Compliance, Revenue Assurance & Governed Case Progression.

12.1 Scope

Covers deterministic calculated Revenue derivation from Traffic and the effective-dated surcharge rate, and its reconciliation against operator-reported Billing figures.

12.2 Principal Components

Table 13. Revenue Intelligence & Billing Verification principal components.

Component	Responsibility
Revenue Calculator	Computes calculated Revenue deterministically from Traffic and the effective-dated surcharge Rule.
Billing Adapter	Ingests operator-reported Billing figures.
Reconciliation Engine	Compares calculated vs reported Revenue and surfaces differences with explicit basis/unit.
Rule Dependency	Reads the current and historical surcharge rate from the Rule Service (Ch.22).

12.3 Calculation Principle

Calculated Revenue and reported Revenue are always kept distinct and separately labelled. The surcharge rate is an effective-dated Rule Service value (currently supplied as USD 0.19) rather than a constant embedded in code, so historical calculations remain correct even after the rate changes.

Figure 13. Revenue calculation and reconciliation relationships.

SOURCE	RELATIONSHIP	TARGET
Traffic Store	supplies volumes to	Revenue Calculator
Rule Service (surcharge rate)	supplies effective-dated rate to	Revenue Calculator
Revenue Calculator	produces calculated Revenue for	Reconciliation Engine
Billing Adapter	supplies reported Revenue to	Reconciliation Engine

12.4 Reconciliation Semantics

A difference between calculated and reported Revenue is surfaced as a difference, not automatically labelled underpayment — it becomes a candidate for the Regulatory Intelligence workflow (Ch.16), where a human determines cause and any enforcement action.

12.5 Data Model

- Calculated Revenue Record — operator, period, traffic basis, applied rate/version, calculated value, unit/currency.
- Reported Revenue Record — operator, period, reported value, source.
- Reconciliation Result — difference, basis, status, linked Case if escalated.

12.6 Security Controls

- Only authorized rate administrators may change the surcharge rate, through the Rule Service change process with effective dating (Ch.22).
- Historical rates are immutable once superseded, preserving past calculations exactly as they were computed.

12.7 Failure Modes

- Rate change pending approval: calculations continue using the last approved effective rate until the new rate's effective date is confirmed.
- Missing Traffic for a period: Revenue for that period is marked not-calculable rather than computed from a zero/default assumption.

12.8 Non-Functional Requirements

- Monthly Revenue calculation and reconciliation for all operators completes within the reporting window required by Regulatory Intelligence.

12.9 Acceptance Criteria

- A surcharge rate change is demonstrably effective-dated: calculations before the change date use the old rate, after use the new rate.
- A reconciliation difference is demonstrably surfaced with basis/unit and does not auto-label as underpayment.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm legal basis, currency/rounding convention and rate-change approval authority with NCA Revenue Assurance and Legal; the current supplied surcharge value of USD 0.19 is treated as the initial Rule Service value pending that confirmation.

Traceability: NCIE-001 §Regulatory Intelligence, Compliance, Revenue Assurance & Governed Case Progression.

Mobile Money Intelligence

Defines approved aggregate Mobile Money intelligence.

Full production content. Traces to NCIE-001 v1.2 Product Capability Architecture.

13.1 Scope

Covers approved aggregate Mobile Money monitoring only — daily sent/received volumes by operator. Subscriber-level Mobile Money data is explicitly out of scope for NCIE-002, matching the NCIE-001 product boundary.

13.2 Principal Components

Table 14. Mobile Money Intelligence principal components.

Component	Responsibility
Mobile Money Adapter	Ingests daily aggregate sent/received figures by operator (Ch.6 channels).
Aggregate Store	Canonical, versioned aggregate records; no subscriber-level fields exist in the schema.
Quality Service	Flags missing days and anomalous swings without auto-correcting.
Analytics Service	Trend and comparison views over aggregate volumes.

13.3 Data Model

- Aggregate Record — operator, date, currency, sent volume, received volume, source, coverage status.

13.4 Security Controls

- The Aggregate Store schema structurally excludes any subscriber-identifying field; this is enforced at the schema level, not merely by access policy, so no future adapter can accidentally introduce subscriber-level data without a deliberate architecture change.

13.5 Failure Modes

- Missing day's figures: remains explicitly missing, never assumed zero, consistent with the platform-wide missing-data principle (Ch.5).

13.6 Non-Functional Requirements

- Daily aggregate ingestion for all operators completes before the start of next-business-day review.

13.7 Acceptance Criteria

- A schema inspection confirms no subscriber-identifying field exists anywhere in the Mobile Money data path.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm source system, currency and daily cut-off time with NCA Mobile Money oversight; confirm whether any additional approved aggregate measures (e.g. transaction counts) should be added — architecture accommodates additional aggregate fields without redesign, subject to the same subscriber-level exclusion.

Traceability: NCIE-001 \$Product Capability Architecture.

SIM Registration Intelligence

Defines protected identity/SIM intelligence and exception classification.

Full production content. Traces to NCIE-001 v1.2 Security, Privacy, Identity, Authorization & Intelligence Sovereignty Governance (amended); Regulatory Intelligence, Compliance, Revenue Assurance & Governed Case Progression.

14.1 Scope

Covers protected identity/SIM registration intelligence, exception classification against configurable registration limits, and governed cross-operator identity resolution.

14.2 Principal Components

Table 15. SIM Registration Intelligence principal components.

Component	Responsibility
Protected Identity Service	Holds Ghana Card and foreign-passport-based identity references under field-level protection (Ch.4).
Registration Store	Canonical SIM registration counts and status per identity/operator.
Identity Resolution Service	Performs governed cross-operator identity matching under heightened audit (Ch.23, Ch.26).
Rule Engine	Applies effective-dated registration-limit rules (current supplied limits: 10 for Ghana Card, 3 for foreign passport).

14.3 Exception Semantics

An identity exceeding the effective registration limit is classified as an exception, not automatically as fraud. Exceptions become candidates for the Regulatory Intelligence workflow (Ch.16), where investigation determines cause (e.g. legitimate multi-line business use) before any enforcement consideration.

14.4 Data Model

- Protected Identity — document class (Ghana Card / foreign passport), protected reference, never displayed in plaintext outside authorized protected-identity views.
- Registration Record — identity reference, operator, SIM count, status.
- Registration Limit Rule — document class, limit value, effective date range (Ch.22).

14.5 Security Controls

- Protected identifiers are field-level access-controlled (Ch.4) and never appear in aggregate or Situational Awareness views by default.
- Cross-operator identity resolution actions are individually audited (Ch.23), given their sensitivity.
- Access to protected identity fields requires MFA and is logged with heightened detail (Ch.26).

14.6 Failure Modes

- Ambiguous identity match across operators: presented for human confirmation rather than auto-merged.

14.7 Non-Functional Requirements

- Registration-limit exception detection runs on each registration-data refresh cycle without manual triggering.

14.8 Acceptance Criteria

- An identity exceeding the effective limit is demonstrably classified as an exception, and not labelled fraud, pending investigation.

- A registration-limit rule change is demonstrably effective-dated, consistent with §12.3's approach to the surcharge rate.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. The current supplied limits of 10 (Ghana Card) and 3 (foreign passport) are treated as the initial Rule Service values; confirm final limits and rule-change authority with NCA registration-policy leadership.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm exact field-level protection scope (which roles may view protected identifiers unmasked) with NCA data-protection/legal leadership before production.

Traceability: NCIE-001 §Security, Privacy, Identity, Authorization & Intelligence Sovereignty Governance; §Regulatory Intelligence, Compliance, Revenue Assurance & Governed Case Progression.

Anti-Fraud / SIMBOX Intelligence

Defines the broader Anti-Fraud domain.

Full production content. Traces to NCIE-001 v1.2 Regulatory Intelligence, Compliance, Revenue Assurance & Governed Case Progression (amended).

15.1 Scope

Covers the broader Anti-Fraud domain: detection, suspicion, investigation, validation, blocking, location, referral and outcome, kept as explicitly distinct stages.

15.2 Principal Components

Table 16. Anti-Fraud / SIMBOX Intelligence principal components.

Component	Responsibility
Anti-Fraud Service	Ingests and correlates detection signals into suspicion candidates.
Investigation Workflow	Governs progression from suspicion through investigation to validated finding.
Location Service	Resolves suspected SIMBOX activity to physical/network location under sensitive-join controls.
Blocking Workflow	Executes approved blocking actions only after validation and authorization.
Evidence Links	Binds every stage to authoritative Evidence objects (Ch.23).

15.3 Stage Model

Figure 14. Anti-Fraud / SIMBOX stage progression, each stage explicit and auditable.

01	Detection (automated signal)
	▼
02	Suspicion (analyst-reviewed candidate)
	▼
03	Investigation (evidence gathering, sensitive joins under audit)
	▼
04	Validation (human-confirmed finding)
	▼
05	Blocking (approved, verified action)
	▼
06	Location & Referral (where applicable)
	▼
07	Outcome (apprehension status — authoritative source only)

15.4 Data Model

- Suspicion Candidate — detection signals, confidence indicators, linked identities/cells.
- Investigation — assigned investigator, evidence links, sensitive-join log.
- Blocking Action — target, authorization reference, execution status, verification result.
- Outcome Record — apprehension status sourced only from the authoritative law-enforcement/regulatory system of record, never inferred by NCIE.

15.5 Security Controls

- Sensitive joins (e.g. identity-to-location) are individually authorized and logged, not a standing capability of the investigator role.

- Blocking actions require both validation and separate authorization approval before execution (separation of duties, Ch.4).
- Blocking is never treated as an admission of guilt; outcome/apprehension status is sourced only from the authoritative system, never asserted by NCIE.

15.6 Failure Modes

- Blocking execution fails at the provider: Blocking Workflow retries under idempotency control and surfaces failure rather than silently reporting success.

15.7 Non-Functional Requirements

- Detection-to-suspicion candidate generation completes within the near-real-time window required for SIMBOX response effectiveness (target confirmed with NCA Anti-Fraud, Ch.30).

15.8 Acceptance Criteria

- Each of the six stages in §15.3 is demonstrably distinct and independently auditable for a representative case.
- A blocking action is demonstrably blocked from execution without both validation and authorization present.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm which Anti-Fraud modules and blocking interfaces are required at launch, and the investigation-stage approval matrix, with NCA Anti-Fraud/SIMBOX leadership.

Traceability: NCIE-001 §Regulatory Intelligence, Compliance, Revenue Assurance & Governed Case Progression.

Regulatory Intelligence, Cases, Findings & Decisions

Defines regulatory workflow and reuse of the existing Regulatory Intelligence platform.

Full production content. Traces to NCIE-001 v1.2 Regulatory Intelligence, Compliance, Revenue Assurance & Governed Case Progression.

16.1 Scope

Defines regulatory case workflow and governs reuse of NCA's existing Regulatory Intelligence platform rather than rebuilding case management from scratch.

16.2 Principal Components

Table 17. Regulatory Intelligence, Cases, Findings & Decisions principal components.

Component	Responsibility
Case Service	Opens and tracks Cases from Exceptions/Candidates across all domains (Traffic, Revenue, MoMo, SIM, Anti-Fraud, Network).
Finding Workflow	Manages Proposed Finding to Validated Finding progression, with AI preparing but never validating.
Decision Service	Records human/institutional Decisions against Validated Findings, including appeal/reversal history.
Evidence Binder	Aggregates all Evidence objects (Ch.23) supporting a Case into a defensible package.
Existing Platform Adapter	Integration boundary with NCA's existing Regulatory Intelligence platform (Ch.25).

16.3 Case Lifecycle

Figure 15. Regulatory Case lifecycle from exception to decision.

01	Exception / Candidate (from any domain)
02	Case opened, linked to Evidence Binder
03	Proposed Finding (AI may assist, never validates)
04	Validated Finding (human-confirmed)
05	Decision (institutional authority)
06	Appeal / Reversal (preserves full history, never overwrites)

16.4 Data Model

- Case — originating domain/exception, status, linked evidence, assigned officer(s).
- Proposed Finding — drafted content, AI-attribution flag where applicable, supporting evidence references.
- Validated Finding — human validator identity, validation date, immutable content.
- Decision — decision content, authority, date, appeal/reversal linkage preserving prior state.

16.5 Security Controls

- The role proposing a Finding is distinct from the role validating it (separation of duties, Ch.4).
- AI-assisted content in a Proposed Finding is explicitly attributed and never silently merged into human-authored text (Ch.23, Ch.24).
- An appeal or reversal preserves the original Decision and Finding history rather than deleting or overwriting it.

16.6 Failure Modes

- Existing Regulatory Intelligence platform unreachable: new Case creation queues locally with Evidence intact; no evidence is lost pending reconnection (Ch.28).

16.7 Non-Functional Requirements

- Case and Evidence Binder assembly for a typical case completes within seconds, independent of the number of linked evidence items up to the platform's stated capacity target (Ch.30).

16.8 Acceptance Criteria

- A Case is demonstrably blocked from reaching Validated Finding status without a distinct validator from the proposer.
- An appealed Decision is demonstrably shown with full prior history intact.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm the precise integration contract (API vs data-sync) with NCA's existing Regulatory Intelligence platform, and confirm case-lifecycle stage names/approval roles, with the Regulatory Intelligence platform owner.

Traceability: NCIE-001 §Regulatory Intelligence, Compliance, Revenue Assurance & Governed Case Progression.

Cross-Domain Fusion & Intelligence Graph

Defines relationships across domains without false causation.

Full production content. Traces to NCIE-001 v1.2 Intelligence Fusion, Cross-Domain Correlation & National Intelligence Synthesis Governance.

17.1 Scope

Defines how Network, Traffic, Revenue, Mobile Money, SIM Registration, Anti-Fraud, Incident, Billing and Regulatory data relate to one another temporally, spatially and by entity — without asserting causation the evidence does not support.

17.2 Principal Components

Table 18. Cross-Domain Fusion & Intelligence Graph principal components.

Component	Responsibility
Fusion Service	Orchestrates cross-domain correlation queries against the canonical platform.
Graph Store	Persists entity relationships and their provenance as a queryable graph.
Entity Resolution	Determines when records from different domains refer to the same real-world entity.
Correlation Engine	Computes temporal/spatial/entity correlations and confidence indicators.
Provenance Graph	Links every fused relationship back to its supporting canonical/evidence records (Ch.23).

17.3 Correlation Principle

The Correlation Engine produces correlation, not causation: a temporal or spatial coincidence between, say, an Incident and a Revenue anomaly is surfaced with its supporting evidence and confidence, and is explicitly labelled as a candidate relationship for human interpretation, never as an asserted cause.

Figure 16. Domain data feeding cross-domain fusion.

SOURCE	RELATIONSHIP	TARGET
Network / Traffic / Revenue Domains	supply canonical facts to	Fusion Service
Mobile Money / SIM / Anti-Fraud Domains	supply canonical facts to	Fusion Service
Fusion Service	writes correlated relationships to	Graph Store
Graph Store	is queried by	Situational Awareness Workspace / Assistant

17.4 Data Model

- Relationship — source entity, target entity, relationship type, confidence, temporal/spatial basis, provenance references.
- Contradiction Record — two relationships or facts that conflict, preserved rather than silently resolved (Ch.5, Ch.23).

17.5 Security Controls

- Fusion queries respect the same field-level authorization as the underlying domains; fusion never bypasses protected-identity controls (Ch.4, Ch.14).

17.6 Failure Modes

- Conflicting cross-domain signals: both retained as a Contradiction Record rather than one overwriting the other.

17.7 Non-Functional Requirements

- Interactive fusion queries for a bounded geography/time window return within 3 seconds at p95 (Ch.30).

17.8 Acceptance Criteria

- A known cross-domain scenario (e.g. Incident correlated with Traffic dip) is demonstrably surfaced as a labelled correlation with supporting evidence, not an asserted cause.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm the priority list of cross-domain fusion use cases for launch with NCIE domain leads; the Fusion Service architecture supports incremental addition of correlation types without redesign.

Traceability: NCIE-001 §Intelligence Fusion, Cross-Domain Correlation & National Intelligence Synthesis Governance.

REWS, Alerts, Situations & National Operating Picture

Defines alerting and multi-domain Situations.

Full production content. Traces to NCIE-001 v1.2 Situational Awareness, Early Warning, Alerting & National Communications Operating Picture.

18.1 Scope

Defines the Regulatory Early Warning System (REWS) engine, Alert lifecycle, multi-domain Situations, and the National Operating Picture that exposes known/missing/stale/waiting/contradictory state at a glance.

18.2 Principal Components

Table 19. REWS, Alerts, Situations & National Operating Picture principal components.

Component	Responsibility
REWS Engine	Evaluates effective-dated alert rules against canonical/analytical data.
Alert Store	Canonical record of raised, acknowledged and closed Alerts.
Situation Service	Groups related Alerts/Incidents/evidence into a multi-domain Situation with its own lifecycle.
Notification Service	Delivers Alert/Situation notifications to subscribed roles.

18.3 Entity Distinctions

Alert, Incident, Situation, Investigation and Case are kept explicitly distinct: an Alert is a rule-driven signal; an Incident is a domain-specific event (Ch.10); a Situation is a human- or rule-initiated grouping of related Alerts/Incidents/evidence across domains; an Investigation is scoped analytical work; a Case is the regulatory workflow container (Ch.16). Promotion between them is explicit, never implicit.

18.4 Alert Lifecycle

Figure 17. Alert lifecycle from rule evaluation to closure.

01	Effective-dated Alert Rule evaluated against canonical/analytical data
02	Alert raised with severity and supporting evidence references
03	Notification Service delivers to subscribed roles
04	Human acknowledges and investigates, or groups into a Situation
05	Alert closed with recorded closure authority and reason

18.5 National Operating Picture States

Table 20. National Operating Picture status vocabulary.

State	Meaning
Known	Data is current and validated for the relevant geography/domain.
Missing	Expected data has not arrived; distinct from a zero or good-status value (Ch.5, Ch.6).
Stale	Data was received but has exceeded its freshness window.
Waiting	A workflow or acquisition Run is in progress (e.g. WAITING_FOR_PM, Ch.21).

State	Meaning
Contradictory	Two sources disagree and the conflict is preserved rather than resolved silently (Ch.17).

18.6 Data Model

- Alert Rule — condition, severity, effective date range, owner.
- Alert — rule reference, raised time, evidence references, acknowledgement/closure trail.
- Situation — grouped Alerts/Incidents/evidence, lifecycle status, assigned owner.

18.7 Security Controls

- Alert closure and Situation-lifecycle changes are audited with actor identity (Ch.23).

18.8 Failure Modes

- REWS Engine degraded: last-known Alert state remains visible with a staleness indicator; the engine fails closed on new rule evaluation rather than silently skipping checks (Ch.28).

18.9 Non-Functional Requirements

- Alert rule evaluation runs on each relevant data refresh cycle with raised-Alert latency bounded per severity tier (Ch.30).

18.10 Acceptance Criteria

- Every National Operating Picture status in §18.5 is demonstrably distinguishable in the workspace for a representative geography.
- A rule-driven Alert is demonstrably traceable to its supporting evidence.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm severity tiers, escalation timing and closure authority per Alert category with NCA operations leadership; the REWS rule model accommodates any agreed severity scheme via the Rule Service (Ch.22).

Traceability: NCIE-001 §Situational Awareness, Early Warning, Alerting & National Communications Operating Picture.

Intelligence Assistant & Conversational Orchestration

Defines the Assistant as a governed NCIE interface.

Full production content, including the v0.4 controlled-update amendment for active AI participation. Traces to NCIE-001 v1.2 NCIE Intelligence Assistant; NCIE Intelligence Assistant, Conversational Intelligence, Collaborative Brainstorming & Governed Memory (amended Ch.14/15/18).

19.1 Scope

Defines the Assistant as a governed NCIE interface: multi-turn conversational access to authoritative platform state, plus — following the v0.4 amendment — the ability to participate proactively in human-to-human discussion under bounded, human-controlled conditions (see also Chapter 20, which holds the primary architecture for collaborative reasoning).

19.2 Principal Components

Table 21. Intelligence Assistant & Conversational Orchestration principal components.

Component	Responsibility
Conversation Service	Manages multi-turn dialogue state per user/room.
Context Resolver	Resolves operator/geography/date/entity context from conversation and workspace state (Ch.3).
Tool Gateway	Exposes evidence/map/rule/source/workflow tools to the model under authorization filtering.
Model Abstraction	Provider-neutral interface to the Model Gateway (Ch.24).
Grounding Layer	Ensures answers are derived from authoritative NCIE state, not model-invented content.
Proactive Contribution Controller	(v0.4) Decides whether/when the Assistant contributes without being directly asked.
Participation Policy Adapter	(v0.4) Applies the active room's Participation Mode (Ch.20 §20.7) to Assistant behaviour.
Collaborative Context Adapter	(v0.4) Feeds the Assistant the current Collaborative Reasoning State (Ch.20) so it follows the live discussion.
Grounded Contribution Planner	(v0.4) Plans a proactive contribution (question, evidence, challenge) grounded in authorized tools before it is offered.
Assistant Naming Configuration	Externalises the user-facing Assistant name (ARGUS) from internal identifiers, APIs, schemas, workflow definitions, audit events and domain services (§19.11).
Agent Capability Discovery Adapter	(v0.5) Determines whether an eligible Registered Reusable Agent adequately satisfies a task before ARGUS requests Dynamic Agent Synthesis via the Agent Factory (Ch.24 §24.4A; §19.2A).

19.2A ARGUS Consumption of Registered and Dynamically Synthesized Agents (v0.5)

ARGUS does not itself construct, validate or register agents; it consumes agent capability through the governed Agent Factory and Agent Registry defined in Chapter 24 §24.4A. When a task requires specialized analytical capability, ARGUS:

- discovers eligible existing Registered Reusable Agents through the Agent Capability Discovery Adapter;
- determines whether an adequate eligible capability already exists;
- where none exists, requests Dynamic Agent Synthesis through the governed Agent Factory;
- receives and uses a bounded Ephemeral Agent created from approved NCIE primitives;
- continues to operate within the same authorization, context, tool and model controls as any other ARGUS-orchestrated capability (§19.7).

ARGUS ≠ AGENT FACTORY: ARGUS requests synthesis; it does not itself perform Agent Definition construction, VPF Gate A validation, sandboxing or activation, all of which remain Agent Factory responsibilities (Ch.24 §24.4A.1). SYNTHESIS ≠ REGISTRATION and SYNTHESIS ≠ AUTHORITY CREATION: a synthesized agent is ephemeral and unregistered by default, and gains no institutional authority ARGUS itself does not already hold. Where a task requires a capability outside the available approved primitives, the architecture routes to an ENGINEERING CHANGE CANDIDATE (Ch.24 §24.4A.1) rather than allowing ARGUS to fabricate infrastructure.

19.3 Invoked Interaction Flow

Figure 18. Standard invoked Assistant interaction.

01	User asks a question in the workspace or a room
02	Context Resolver grounds operator/geography/date/entity context
03	Tool Gateway invokes authorized evidence/map/rule/source/workflow tools
04	Grounding Layer composes an answer from authoritative results only
05	Conversation Service returns the answer with source references

19.4 Proactive Contribution (v0.4 amendment)

In Participatory Mode (Ch.20 §20.7), the Assistant is architecturally permitted to contribute without a direct question, subject to the following required behaviours:

- Follows permitted human-to-human discussion and resolves operator, geography, time, cells, hypotheses and the active analytical thread before contributing.
- May ask analytically useful questions, introduce authorized evidence, challenge assumptions, compare hypotheses and identify evidence gaps.
- Every proactive contribution is grounded in authoritative NCIE services via the Tool Gateway — never in unexamined model reasoning alone.
- Observes non-dominance: useful, occasional contribution, not a reply after every human message; humans can sustain uninterrupted human-to-human discussion.

Example invoked query: “ARGUS, why is this cell degraded?” Example proactive contribution, always attributed and never presented as confirmed fact: “ARGUS Hypothesis — Possible Shared Transmission Dependency,” not “Confirmed Cause — Shared Transmission Dependency.”

19.5 Participation Modes Consumed

The Assistant does not own participation-mode state; it consumes the mode set at the room level by the Participation Mode Manager (Ch.20 §20.7) through the Participation Policy Adapter.

19.6 Data Model

- Conversation — participant(s), room reference (if any), turn history, resolved context.
- Tool Invocation — tool name, parameters, authorization check result, response reference.
- Proactive Contribution Candidate — trigger, planned content, grounding references, offered/suppressed decision.

19.7 Security Controls

- Tool Gateway enforces the same authorization layers as direct API access (Ch.4); the Assistant never has implicit elevated access.
- Proactive contributions never introduce evidence the current room participants are not individually authorized to see (Ch.20 §20.9).
- (v0.5) Dynamic Agent Synthesis does not grant ARGUS or the synthesized Ephemeral Agent elevated access beyond the requesting task’s current authorization; the agent remains subject to the same Model Gateway, Tool Gateway and Context/Memory boundaries as invoked Assistant use (Ch.24 §24.4A.1, §24.4A.8).

19.8 Failure Modes

- Model Gateway unavailable: Conversation Service degrades to direct tool/query access without conversational framing; core NCIE data remains reachable (Ch.28).
- Grounding Layer cannot verify a claim against authoritative state: the Assistant states the limitation rather than presenting an ungrounded answer as fact.
- (v0.5) Agent Factory or VPF unavailable: ARGUS continues to operate using eligible Registered Agents and direct governed model use; Dynamic Agent Synthesis and validated release of new substantive AI output degrade explicitly rather than silently (Ch.28 §28.3).

19.9 Non-Functional Requirements

- Invoked responses grounded in a single tool call return within 3 seconds at p95 (Ch.30).
- Proactive contribution evaluation does not add perceptible latency to ordinary human-to-human messages in a room.

19.10 Acceptance Criteria

- An invoked question is demonstrably answered using only authoritative tool results, with sources shown.
- In Participatory Mode, the Assistant demonstrably refrains from responding to every message (non-dominance) while still surfacing at least one genuinely useful unprompted contribution in a scripted test discussion.

19.11 ARGUS Naming Boundary

The NCIE Intelligence Assistant is presented to users under the institutional pseudo-name ARGUS — Adaptive Regulatory Governance & Unified Surveillance. This section fixes the boundary between that presentation identity and the underlying architecture defined by this chapter and Chapter 20.

- ARGUS = User-Facing Assistant Identity, and not ARGUS = Separate Intelligence System — ARGUS is a human-friendly invocation and presentation identity for the one NCIE Intelligence Assistant described throughout this specification, never a second, independently governed system.
- Configurable naming — the user-facing name is externalised through the Assistant Naming Configuration component (§19.2) and is not hard-coded into APIs, database schemas, workflow definitions, audit events or domain services, so the institutional name can change without redesigning the Assistant architecture.
- Stable identity in audit — regardless of the user-facing name in effect, material Assistant actions retain a stable machine/service identity for audit (Ch.23), provenance, security (Ch.4, Ch.26) and model governance (Ch.24).
- Authorization binds to the stable internal identity, not the display name — whether presented as ARGUS or a future alternate institutional name, the Authorization Service (Ch.4) and Model Registry (Ch.24) evaluate the same underlying service identity, so a rename never requires reissuing policy, approvals or evaluation history.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm which conversational actions require explicit user confirmation before execution (e.g. triggering an acquisition workflow via Ch.21) versus which are safe to answer directly; default is that any action with a side effect requires confirmation, and any read-only query does not.

Traceability: NCIE-001 §NCIE Intelligence Assistant; §NCIE Intelligence Assistant, Conversational Intelligence, Collaborative Brainstorming & Governed Memory; NCIE-001 v1.3 §12.77-§12.79; NCIE-007 v1.1 Ch.11 §11.3, §11.9.

Memory, Collaborative Intelligence & Multi-User Brainstorming

Defines persistent memory, permission-aware multi-user brainstorming, and the architecture that allows the NCIE Intelligence Assistant to participate as a governed, non-dominant collaborator.

Full production content — v0.4 in-place rewrite. This is the primary architecture for Human ↔ Human ↔ NCIE collaborative reasoning. Traces to NCIE-001 v1.2 Evidence, Knowledge, Memory & Provenance Governance; NCIE Intelligence Assistant, Conversational Intelligence, Collaborative Brainstorming & Governed Memory (amended Ch.14/15/18).

20.1 Scope

Defines persistent private/shared/institutional memory, and the architecture that allows NCIE to participate as a governed, non-dominant collaborator in multi-user brainstorming rather than a passive, invoked-only chatbot. The interaction model is Human ↔ Human ↔ NCIE: humans retain intellectual direction and institutional authority at all times.

20.2 Memory Tiers

Table 22. NCIE memory tiers and sharing rules.

Tier	Description	Sharing Rule
Private User Memory	Persistent per-user memory of preferences, working context and history.	Never automatically shared, even into a room the user joins.
Shared Room Memory	Memory scoped to a specific collaborative room.	Visible to current, authorized room participants only.
Institutional Memory	Formally promoted knowledge (validated Findings, Decisions, adopted conclusions).	Governed by the same authorization rules as its source domain.

20.3 Active AI Collaborative Participation

The following capabilities are explicit architectural requirements, not implied side effects of giving a room shared chatbot access:

- Active AI Brainstorming Participation — NCIE can participate in the intellectual exchange rather than merely wait for direct questions.
- AI Questioning — in Participatory Mode, NCIE may proactively ask analytically useful questions exposing assumptions, missing information, unexplored dimensions or reasoning weaknesses.
- AI Hypothesis Generation — NCIE may introduce clearly AI-attributed alternative hypotheses for human consideration; these remain working reasoning, not findings.
- AI Challenge Function — NCIE may challenge human or AI hypotheses using authorized evidence, inconsistencies, temporal or spatial relationships, source limitations and credible alternative explanations.
- Contradiction Seeking — NCIE may actively seek authorized evidence that contradicts the dominant working hypothesis, testing confirmation bias rather than reinforcing consensus.
- Competing-Hypothesis Analysis — multiple human and AI hypotheses may coexist, compared against supporting evidence, contradictory evidence, temporal fit, spatial fit, source quality, coverage and unresolved gaps.
- Discriminating-Evidence Identification — NCIE may identify which additional evidence would most help distinguish competing explanations.
- Evidence-Gap Identification — NCIE may identify missing PM, topology, incident, Traffic, QoS Campaign or other authorized evidence relevant to the discussion.

Every AI-originated hypothesis and question is rendered under ARGUS's name with explicit AI attribution (e.g. “ARGUS Hypothesis — Possible Shared Transmission Dependency”), and is never rendered in a form that could be mistaken for a confirmed cause or a human-originated Finding (Ch.19 §19.4).

20.3A Realization via Registered or Ephemeral Synthesized Agents (v0.5)

Where a collaborative analytical role in §20.3 is realized using an AI agent rather than ARGUS's own direct model use, it may be realized through an approved Registered Reusable Agent or a bounded Ephemeral Synthesized Agent created under NCIE-007 governance (Ch.24 §24.4A). Such agents remain subordinate analytical capabilities: they do not gain Room authority, do not inherit pooled participant permissions (§20.9), do not become institutional decision makers, and do not change ARGUS's human-facing role or naming (Ch.19 §19.11).

20.4 Principal Components

Table 23. Memory, Collaborative Intelligence & Multi-User Brainstorming principal components.

Component	Responsibility
Collaborative Reasoning State Service	Owns the Collaborative Reasoning State: distinct from raw transcript and memory.
Collaborative Context Manager	Tracks current focus, active analytical thread and room membership context.
Multi-Speaker Attribution Service	Preserves who introduced, challenged, adopted, modified or rejected each idea, including explicit AI origin.
Analytical Thread Manager	Keeps multiple concurrent questions/threads from cross-contaminating evidence or hypotheses.
Participation Mode Manager	Owns and enforces the room's current Participation Mode.
Proactive Contribution Controller	Shared with Ch.19; gates whether a planned AI contribution is actually offered.
Hypothesis Lifecycle Manager	Tracks human/AI hypotheses from proposal through comparison to adoption, deferral or rejection.
Contradiction / Evidence-Gap Service	Surfaces contradictions and evidence gaps relevant to the active thread.
Shared Evidence Board	Organizes references to authoritative Evidence objects, maps, campaigns, calculations, hypotheses, contradictions, notes and tasks without replacing the Evidence Service (Ch.23).
Room-to-Workflow Bridge	Shared with Ch.21; carries room-originated acquisition suggestions to the Automation Orchestrator.
Room Summary & Shift-Handover Service	Reconstructs 'where were we?' by combining Shared Room Memory, Collaborative Reasoning State and current authoritative NCIE state.
Memory Promotion Service	Governs deliberate promotion of collaborative reasoning into Institutional Knowledge, Findings or Decisions.

20.5 Collaborative Reasoning State

The Collaborative Reasoning State is a distinct object from the raw conversation transcript and from memory. It tracks: current focus, analytical threads, human hypotheses, AI hypotheses, evidence, contradictions, gaps, rejected/deferred ideas, tasks, automation requests and evidence being waited on. It is what the Room Summary & Shift-Handover Service reconstructs for a returning or newly joined participant.

20.6 Multi-Speaker Attribution & Analytical Threads

- Every contribution — human or AI — retains its originator, and adoption by a human never erases AI origin.
- Multiple analytical threads can coexist in one room without evidence or hypotheses being silently mixed between them.
- Room membership changes trigger authorization-aware handling of historical context — a newly joined participant does not automatically see history predating authorization checks passing for them.

20.7 Participation Modes

Table 24. Persistent, human-controlled room Participation Modes.

Mode	Behaviour
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Mode	Behaviour
Participatory	Assistant may proactively contribute per §20.3, subject to non-dominance.
Invoked	Assistant responds only when directly addressed; no proactive contribution.
Evidence-Watch	Assistant stays silent on discussion but may surface material new evidence events (Ch.20 §20.10) as queued room events.
Silent	Assistant takes no conversational action; room is human-only until mode changes.

The active mode is always visible in the workspace under ARGUS's user-facing name, e.g. “ARGUS — Participatory,” “ARGUS — Invoked,” “ARGUS — Evidence-Watch,” “ARGUS — Silent” (Ch.19 §19.11). Authorized users may change the mode through structured UI controls and, where permitted, natural-language commands such as “ARGUS, don't interrupt unless new evidence arrives.”

Humans may also invoke Specialized Participation Roles — Devil's Advocate, Evidence Challenger, Historical Comparator, Data-Gap Analyst — which shape how NCIE contributes within Participatory Mode without granting any additional authority or data access beyond the participant's own permissions. In this configuration the workspace shows, for example, “ARGUS — Participatory / Evidence Challenger.”

20.8 Non-Dominance and Human Intellectual Control

- Participatory Mode means useful proactive contribution, not a reply after every human message; humans must be able to sustain uninterrupted human-to-human discussion.
- Humans may accept, reject, modify, defer, redirect or formally promote any AI contribution, and may increase, constrain or silence AI participation at any time.

20.9 Permission-Aware Collaboration

- Shared room context never pools participant permissions; each participant sees only what their own authorization permits, even within one room (Ch.4).
- Private User Memory is not automatically shareable merely because it would be relevant to the room's discussion — sharing requires a deliberate user action.

20.10 Cross-Session Continuity and Event Awareness

- ‘Where were we?’ reconstruction combines Shared Room Memory, Collaborative Reasoning State and current authoritative NCIE state — never a stale snapshot presented as current.
- Validated PM arrival, workflow completion, new incident reports, Traffic arrival or source corrections may return to the room as events in Participatory/Evidence-Watch mode.

20.11 Room-to-Workflow Integration

Brainstorming may lead to governed evidence-acquisition requests via the Room-to-Workflow Bridge (shared with Ch.21). AI suggestion of an acquisition action remains architecturally distinct from execution authority; validated results return asynchronously to the originating Collaborative Reasoning State.

20.12 Formal Promotion Boundary

Collaborative reasoning remains distinct from formal analysis, Institutional Knowledge, Findings and Decisions until deliberately promoted through the Memory Promotion Service, which applies the same governance as the target object type (e.g. a promoted hypothesis becoming a Proposed Finding follows Ch.16's Finding Workflow).

20.13 Cross-Chapter Architectural Invariants

The following invariants, introduced by the v0.4 collaborative-brainstorming amendment, bind Chapters 19, 20, 21, 23 and 24 collectively:

- Active AI participation does not create institutional authority.

- Collaborative Reasoning State is distinct from transcript, memory, evidence, formal analysis, finding and decision.
- Human adoption of an AI idea does not erase AI origin.
- Shared room context never pools participant permissions.
- Private memory is not shareable merely because it is relevant.
- Participatory Mode means useful proactive contribution, not continuous AI speech.
- Evidence-Watch does not create unrestricted monitoring authority.
- AI-suggested acquisition is not execution authority.
- Validated workflow results may return to the originating room asynchronously.
- Previous collaborative state must be reconciled with current authoritative NCIE state.
- Multiple analytical threads must remain separable.
- AI challenge is advisory; humans retain intellectual direction and institutional authority.
- (v0.5) A Registered Reusable Agent or Ephemeral Synthesized Agent realizing a collaborative role gains no Room authority, pooled permissions or institutional decision authority by virtue of participating.

20.14 Data Model

- Room — participants, permission map, Participation Mode, linked threads.
- Analytical Thread — topic, linked hypotheses, linked evidence, status.
- Hypothesis — originator (human/AI), content, supporting/contradicting evidence links, status (active/adopted/deferred/rejected).
- Collaborative Evidence Board Entry — reference to authoritative Evidence, added-by, thread linkage.
- Promotion Record — source collaborative object, target formal object, promoting actor, timestamp.

20.15 Security Controls

- Room join/leave and permission changes are enforced by the Authorization Service (Ch.4), not by room-local logic.
- Every promotion action is audited with full before/after state (Ch.23 §23.9).

20.16 Failure Modes

- Model Gateway unavailable during a room session: room continues as human-only (Silent-equivalent behaviour) without losing Collaborative Reasoning State (Ch.28).
- Conflicting concurrent edits to the Collaborative Reasoning State: resolved with last-writer-wins per field plus a visible edit trail, never silent data loss.

20.17 Non-Functional Requirements

- Room Summary/Shift-Handover reconstruction for a typical room completes within 5 seconds at p95.

20.18 Acceptance Criteria

- A scripted two-human, mixed-permission brainstorming session demonstrates permission-aware filtering, non-dominant proactive contribution, and correct multi-speaker attribution.
- A promoted hypothesis is demonstrably traceable from its originating Collaborative Reasoning State entry to the resulting Proposed Finding (Ch.16).

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm the default Participation Mode for ordinary rooms (proposed default: Participatory) versus Anti-Fraud/regulatory rooms (proposed default: Invoked, given sensitivity), and who may change mode (proposed default: room owner or any participant with Case-officer-equivalent role), with NCIE governance leadership.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm which specialized brainstorming roles (§20.7) are required at launch versus a later phase; proposed default is Devil's Advocate and Evidence Challenger at launch, with Historical Comparator and Data-Gap Analyst following in a subsequent release.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm room/thread retention period; proposed default aligns with the Evidence retention period in Ch.23 given rooms may contain material analytical content.

Traceability: NCIE-001 §Evidence, Knowledge, Memory & Provenance Governance; §NCIE Intelligence Assistant, Conversational Intelligence, Collaborative Brainstorming & Governed Memory (Ch.14/15/18 amendments); NCIE-007 v1.1 Ch.15 (Multi-Agent Coordination & Conflict Handling).

Automation Orchestrator, RPA, Scheduling & Persistent Workflows

Defines provider-neutral automation.

Full production content, including the v0.4 room-to-workflow amendment. Traces to NCIE-001 v1.2 Automation Orchestration & RPA Data Acquisition; Automation Orchestration, RPA Governance, Scheduling & Autonomous Workflow Execution.

21.1 Scope

Defines provider-neutral automation: the Capability Registry abstraction, Request/Run state machine, recurring schedules, persistent wait states, and the governed bridge that lets collaborative rooms (Ch.20) suggest and track acquisition workflows.

21.2 Principal Components

Table 25. Automation Orchestrator, RPA, Scheduling & Persistent Workflows principal components.

Component	Responsibility
Capability Registry	Abstracts UiPath, Power Automate, Automation Anywhere and future providers behind one capability contract.
Orchestrator	Dispatches Requests to providers, tracks Runs, applies failover/retry/idempotency.
Scheduler	Manages recurring schedules and condition-based triggers.
Workflow Engine	Executes persistent multi-step workflows, including long-lived WAITING_FOR_PM-style states.
Provider Adapters	Provider-specific execution shims behind the Capability Registry contract.
Human Tasks	Represents steps requiring human action within an otherwise automated workflow.
Room-to-Workflow Bridge	(v0.4, shared with Ch.20) Carries room-originated Automation Requests with room/thread provenance.
Workflow-to-Room Event Adapter	(v0.4) Returns validated workflow results asynchronously into the originating Collaborative Reasoning State.
Collaborative Request Correlation	(v0.4) Correlates a room-suggested request with its resulting Request/Run/Result chain.

21.3 Request/Run State Model

Request, Run, provider, bot and validation states are kept explicitly separate so a failed Run does not corrupt the Request's history and a provider substitution does not require redefining the Request.

Figure 19. Automation Request lifecycle from trigger to delivered result.

01	Automation Request (scheduled, event-triggered, or room-suggested)
02	Capability Registry selects provider/bot
03	Orchestrator dispatches Run with idempotency key
04	Provider executes; Workflow Engine tracks WAITING_FOR_* states as needed
05	Validation Pipeline checks Run output (Ch.6)
06	Result delivered to originating consumer (workspace, schedule, or Room via Workflow-to-Room Event Adapter)

21.4 Room-Originated Requests (v0.4)

- Collaborative rooms can suggest and initiate governed evidence-acquisition workflows via the Room-to-Workflow Bridge.
- AI suggestion of an acquisition remains architecturally distinct from execution authority — a human authorizes execution (Ch.4 separation of duties).
- Room-originated Automation Requests retain room/thread provenance for later reconstruction (Ch.20 §20.5).
- Validated results return asynchronously into the originating Collaborative Reasoning State via the Workflow-to-Room Event Adapter.
- WAITING_FOR_PM, WAITING_FOR_TRAFFIC and similar persistent states can be watched by a room in Evidence-Watch mode (Ch.20 §20.7) without granting the room broader monitoring authority.
- High-risk actions remain behind human authorization/approval even when proposed during brainstorming.

21.5 Data Model

- Automation Request — trigger type (schedule/event/room), target capability, parameters, authorization state.
- Run — provider, bot identity, start/end, status, idempotency key, output artifact reference.
- Schedule — recurrence rule, target Request template, owner.
- Persistent Wait State — named state (e.g. WAITING_FOR_PM), entered/resolved timestamps, watching rooms.

21.6 Security Controls

- Provider credentials are held by the Machine Identity Registry (Ch.4), never embedded in workflow definitions.
- Room-suggested high-risk Requests require explicit human authorization distinct from the suggesting AI or the requesting room membership alone.

21.7 Failure Modes

- Provider outage: Orchestrator fails over to an alternate configured provider where available, or queues the Request with visible degraded status; waiting workflows survive a platform restart (Ch.28).
- Duplicate trigger (e.g. re-delivered event): idempotency key prevents a duplicate Run from executing twice.

21.8 Non-Functional Requirements

- Scheduled Requests fire within 1 minute of their scheduled time at p99.
- Failover to an alternate provider, where configured, completes within the provider's own timeout plus a bounded retry window (Ch.30).

21.9 Acceptance Criteria

- A room-suggested acquisition Request is demonstrably blocked from execution without separate human authorization, and its result demonstrably returns to the originating room.
- A simulated provider outage demonstrably triggers failover or a visibly queued, non-silent degraded state.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm auto-escalation-versus-approval policy per automation category, and RPA provider policy (which providers are approved and for which capability classes), with NCIE automation governance leadership.

Traceability: NCIE-001 §Automation Orchestration & RPA Data Acquisition; §Automation Orchestration, RPA Governance, Scheduling & Autonomous Workflow Execution.

Rules, Calculation, Configuration & Temporal Governance

Centralizes changeable rules and deterministic calculations.

Full production content. Traces to NCIE-001 v1.2 Regulatory Intelligence, Compliance, Revenue Assurance & Governed Case Progression; Governance, Auditability, Explainability, Human Oversight, Operational Resilience & Product Acceptance.

22.1 Scope

Centralizes every changeable rule and deterministic calculation used elsewhere in this specification — the surcharge rate (Ch.12), SIM registration limits (Ch.14), KPI thresholds (Ch.7) and REWS alert rules (Ch.18) — so that none of it is hardcoded inside AI or bot logic.

22.2 Principal Components

Table 26. Rules, Calculation, Configuration & Temporal Governance principal components.

Component	Responsibility
Rule Service	Authoritative store of effective-dated rules consumed by all domain calculators.
Configuration Service	Non-rule operational configuration (thresholds unrelated to calculation, feature toggles).
Calculation Engine	Executes deterministic calculations against current or historical rule versions.
Version Store	Retains every historical and future-scheduled rule version, immutable once superseded.

22.3 Temporal Governance Principle

Rules are effective-dated: a rule has a defined start (and optionally end) date. Both historical and future-scheduled rules can exist simultaneously. A calculation always uses the rule version effective at the calculation's reference date, not the rule version current at query time — this is what makes decision-time reconstruction (Ch.23) correct.

22.4 Conflict Handling

Two rule versions with overlapping effective ranges for the same rule key are a configuration error: the Rule Service blocks authoritative output for the affected calculation rather than silently picking one, and surfaces the conflict for administrator resolution.

22.5 Scenario Modelling

“What-if” scenario runs (e.g. modelling a proposed rate change) use a scenario-scoped rule overlay and never overwrite the authoritative Version Store history.

22.6 Data Model

- Rule — key, value, effective start/end date, version, approving actor.
- Rule Change Request — proposed value, proposed effective date, approval status.
- Scenario — named what-if run, overlay rules, isolated from authoritative history.

22.7 Security Controls

- Rule changes require an authorized rule administrator and are individually audited (Ch.23).
- AI and bot components read rules through the Rule Service API only; no chapter of this specification permits a rule value to be embedded in model prompts or bot scripts as a constant.

22.8 Failure Modes

- Rule Service unavailable: Calculation Engine uses the last successfully cached rule set with a visible staleness indicator; it does not fall back to a hardcoded default (Ch.28).

22.9 Non-Functional Requirements

- Rule lookups for calculation complete within 50ms at p95 given caching (Ch.30).

22.10 Acceptance Criteria

- An overlapping-effective-date rule conflict is demonstrably blocked rather than silently resolved.
- A scenario run is demonstrably isolated from and does not alter authoritative rule history.

Traceability: NCIE-001 §Regulatory Intelligence, Compliance, Revenue Assurance & Governed Case Progression; §Governance, Auditability, Explainability, Human Oversight, Operational Resilience & Product Acceptance.

Evidence, Provenance, Audit & Decision-Time Reconstruction

Defines defensibility from source to decision.

Full production content, including the v0.4 collaborative-provenance amendment. Traces to NCIE-001 v1.2 Evidence, Knowledge, Memory & Provenance Governance.

23.1 Scope

Defines defensibility from source to decision: end-to-end lineage, attribution, contradiction preservation, decision-time reconstruction, and — following the v0.4 amendment — provenance for collaborative human/AI reasoning.

23.2 Principal Components

Table 27. Evidence, Provenance, Audit & Decision-Time Reconstruction principal components.

Component	Responsibility
Evidence Service	Authoritative store of Evidence objects with source, acquisition and validation lineage.
Provenance Graph	Links every derived fact/finding back through its transformation chain to origin evidence.
Audit Service	Records actor (human/AI/bot/service), action, target and timestamp for sensitive reads/actions.
Historical Snapshot Service	Reconstructs authoritative state as it existed at a prior point in time.
Collaborative Provenance Extension	(v0.4) Extends provenance to human/AI contribution origin within collaborative rooms.
Promotion Provenance	(v0.4) Tracks the path: AI/Human contribution → working hypothesis → formal analytical object.
Material Collaboration Audit Events	(v0.4) Audits participation-mode changes, sensitive evidence introduction, formal promotion and room-originated automation.

23.3 End-to-End Lineage

Every canonical fact, calculation, Finding and Decision carries a chain back to its originating Evidence: source system, acquisition Run (Ch.6), applicable rule version (Ch.22), and every transformation applied. Human, AI, bot and service attribution is recorded at each step.

Figure 20. End-to-end provenance chain from evidence to decision.

SOURCE	RELATIONSHIP	TARGET
Source Evidence	supports	Canonical Fact
Canonical Fact + Rule Version	produces	Calculation / Correlation
Calculation / Correlation	supports	Proposed Finding
Proposed Finding	becomes (on human validation)	Validated Finding / Decision

23.4 Contradiction Preservation

Contradictory evidence or facts are preserved side by side with their provenance (Ch.5, Ch.17), never silently reconciled by discarding one. A Decision made despite a known contradiction records that the contradiction existed and was considered.

23.5 Decision-Time Reconstruction

The Historical Snapshot Service can reconstruct exactly what evidence, rule versions and analytical objects were available at a given decision's time, explicitly excluding evidence that arrived later — this is what makes a Decision defensible against later-discovered information.

23.6 Collaborative Provenance (v0.4)

- Human and AI contribution origin remains visible when collaborative reasoning (Ch.20) becomes material to a Finding or Decision.
- Promotion Provenance tracks the full path from an AI or human contribution through working hypothesis to formal analytical object.
- The Collaborative Evidence Board (Ch.20 §20.4) references authoritative evidence rather than copying authority into the room — the room is never itself a source of truth.
- Material collaborative events are audited: participation-mode changes, introduction of sensitive evidence, formal promotion, and room-originated automation (Ch.21 §21.4).
- Private model chain-of-thought is not required to be stored or disclosed; provenance covers grounded tool inputs/outputs and stated conclusions, not internal model deliberation.
- Decision-time reconstruction can show which collaborative evidence and formal objects were available at the time, alongside the standard evidence chain.

23.7 Data Model

- Evidence — content reference, source, acquisition metadata, validation status, classification.
- Provenance Edge — from-object, to-object, transformation type, actor (human/AI/bot/service), timestamp.
- Audit Event — actor, action, target, timestamp, sensitivity tier.
- Snapshot Request — reference date/time, resulting reconstructed state reference.

23.8 Security Controls

- Sensitive reads (protected identities, Anti-Fraud sensitive joins) and sensitive actions (blocking, rule changes) are always audited, never optional (Ch.14, Ch.15, Ch.22).
- Audit records are append-only and themselves covered by the platform's backup/DR posture (Ch.28).

23.9 Failure Modes

- Audit Service degraded: sensitive actions that require audit are blocked from completing rather than proceeding unaudited (fail closed).

23.10 Non-Functional Requirements

- A decision-time reconstruction for a typical Case completes within 10 seconds at p95 (Ch.30).

23.11 Acceptance Criteria

- A Finding's full provenance chain from source evidence to Decision is demonstrably reconstructable.
- A decision-time reconstruction demonstrably excludes evidence that arrived after the decision's timestamp.
- A promoted collaborative hypothesis is demonstrably traceable via Promotion Provenance to its originating room contribution.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm Evidence and audit retention periods, and the specific list of heightened-audit event types beyond those already named in this chapter, with NCA audit/compliance leadership.

Traceability: NCIE-001 §Evidence, Knowledge, Memory & Provenance Governance.

AI Platform, Model Gateway, Guardrails & Evaluation

Defines AI integration without giving the model ownership of NCIE state.

Full production content, including the v0.4 collaborative-evaluation amendment. Traces to NCIE-001 v1.2 NCIE Intelligence Assistant; Security, Privacy, Identity, Authorization & Intelligence Sovereignty Governance.

24.1 Scope

Defines AI integration without giving any model ownership of NCIE state: provider abstraction, minimal authorized context, guardrails, and — following the v0.4 amendment — evaluation specific to active collaborative participation before it is enabled in production.

24.2 Principal Components

Table 28. AI Platform, Model Gateway, Guardrails & Evaluation principal components.

Component	Responsibility
Model Gateway	Provider-neutral abstraction over internal/external AI models.
Context Builder	Assembles authorization-filtered, minimal context for a given request.
Policy / Guardrails	Enforces prompt-injection defense, secrets exclusion and output policy.
Model Registry	Catalogue of approved models/providers with data-handling classification.
Evaluation Harness	Runs NCIE-specific evaluations before a model/provider is approved for production use.
Collaborative Evaluation Harness	(v0.4) Evaluates collaborative-specific behaviour before Participatory Mode production use.
Participation Guardrail Policy	(v0.4) Prevents permission pooling, private-memory leakage and privilege escalation during proactive participation.
Context Minimizer	(v0.4) Reduces room context sent to external models to the minimum necessary.
Collaborative Quality Monitor	(v0.4) Monitors over-participation and under-participation as distinct, tracked failure modes.
Agent Factory	(v0.5) Governed construction/preparation of bounded Agent Definitions from approved primitives (§24.4A.1).
Agent Registry / Capability Registry	(v0.5) Distinguishes Registered Reusable Agents from Ephemeral Agent Definitions/Instances and Promotion Candidates (§24.4A.2).
Ephemeral Agent Runtime	(v0.5) Bounded execution environment for run-scoped task-specific agents (§24.4A.3).
VPF	(v0.5) Governed validation architecture for substantive AI-generated output, comprising Gate A, Gate B and the VPF Profile Registry (§24.4A.4-§24.4A.7).

24.3 Context and Provider Boundary

The Context Builder assembles only the authorization-filtered, minimal context a request needs. No secret, credential or field the requesting identity is not authorized to see crosses the Model Gateway boundary. External model use is governed by explicit data-residency and retention/training policy per model in the Model Registry.

Figure 21. AI Platform request path from requesting service to model, under Context Builder and Guardrails control.

SOURCE	RELATIONSHIP	TARGET
Requesting Service (Assistant, Room, Workflow)	requests context from	Context Builder

SOURCE	RELATIONSHIP	TARGET
Context Builder	applies authorization filter, sends minimal context to	Model Gateway
Model Gateway	routes to	Approved Internal/External Model
Policy / Guardrails	inspects request and response of	Model Gateway

24.4 Guardrails

- Prompt-injection defense: content originating from ingested sources (email, files, external text) is treated as data, never as instructions to the model.
- No secrets in prompts: credentials, API keys and similar are structurally excluded from any context assembled by the Context Builder.
- External retention/training use is governed per model in the Model Registry; default is training-use disabled unless explicitly approved.
- Participation Guardrail Policy (v0.4) specifically prevents a proactive contribution from pooling room permissions, leaking Private User Memory, or escalating privilege beyond the contributing participant's own access.

24.4A AI Orchestration Extension — Agent Factory, Ephemeral Agents & VPF (v0.5)

This section extends, and does not replace, the AI Platform architecture in §24.2-§24.4. ARGUS continues to operate through the governed Model Gateway and Tool Gateway boundaries with no implicit elevated access; authorization remains current and externally governed; deterministic NCIE capabilities continue when AI is unavailable; and AI does not approve institutional Decisions or production state. Authority: NCIE-001 v1.3 §12.77-§12.80A, §18.241A; NCIE-007 v1.1. This section names no specific Agent Framework, container runtime, model provider, sandbox product, workflow engine, VPF software or registry product as architectural truth — NCIE-004 governs technology selection (§1.3.2).

24.4A.1 Agent Factory

Responsible for governed construction and preparation of bounded task-specific Agent Definitions from approved primitives. It may coordinate Definition Construction, Structural Validation, Policy Compilation, VPF Gate A (§24.4A.5), Sandbox/Evaluation, Activation Preparation and Lifecycle Integration. The Agent Factory is a logical capability, not a requirement for one physical microservice (NCIE-004 governs implementation). CODEX / ENGINEERING BUILDS AND MAINTAINS THE AGENT FACTORY AND INTRODUCES NEW ENGINEERING PRIMITIVES; ARGUS USES THE FACTORY TO SYNTHESIZE BOUNDED AGENTS FROM APPROVED PRIMITIVES. Codex is therefore not a runtime dependency for each individual bounded agent; Codex/engineering re-enters only when a task requires a new Tool, Connector, Runtime Capability, Security Primitive, Persistent Service, Infrastructure or material Production Engineering Change (§24.4A.1 boundary; Ch.32 §32.4A).

24.4A.2 Agent Registry / Capability Registry

Recognizes the governed reusable-agent registry, distinguishing Registered Reusable Agents from Ephemeral Agent Definitions/Instances and from Promotion Candidates. Not every Ephemeral Agent becomes a permanent Registry entry; promotion is a separate, governed lifecycle action (NCIE-007 v1.1 Ch.12, Ch.33).

24.4A.3 Ephemeral Agent Runtime

A logical execution boundary for run-scoped, task-specific agents. The runtime enforces Task Scope, Current Authorization, Permitted Models, Permitted Tools, Permitted Data/Context, Resource Limits, Network/Egress, Lifetime and Termination for every Ephemeral Agent it hosts.

24.4A.4 VPF

The governed validation architecture for substantive AI-generated output. ALL SUBSTANTIVE AI-GENERATED OUTPUT SHALL ENTER THE APPLICABLE VPF GOVERNANCE PATH BEFORE RELEASE AS A VALIDATED NCIE AI RESULT. Validation rigor may vary by governed profile, risk and output class (§24.4A.7); purely deterministic, non-AI processing remains outside the AI-output VPF mandate. VPF FAILURE SHALL NOT FAIL OPEN (Ch.28 §28.3).

24.4A.5 VPF Gate A

Architecturally placed before novel Ephemeral Agent activation. Gate A validates the synthesized agent's capability envelope before execution. It does not validate future agent output — that is Gate B's responsibility (§24.4A.6).

24.4A.6 VPF Gate B

Architecturally placed after substantive AI generation and before validated release or persistence. Gate B applies to substantive output from ARGUS direct model use, Registered Agents, Ephemeral Agents and multi-agent synthesis alike — no orchestration path is exempt by default.

24.4A.7 VPF Profile Registry

Recognizes logical governance for VPF Profile, Profile Version, Applicable Task/Risk/Output Class, Lifecycle State and Current Eligibility. Detailed profile schema is not defined here; NCIE-007 v1.1 Ch.26 §26.5 remains authoritative.

24.4A.8 Orchestration Control Plane and Agent Execution Plane

This chapter distinguishes two logical planes. The Orchestration Control Plane is responsible for Agent Discovery, Synthesis Request, Lifecycle, Registry State, Eligibility, Policy, Risk/Activation Requirements, VPF Requirements, Suspension and Routing. The Agent Execution Plane executes the bounded runtime workload only.

Figure 22. Control Plane / Agent Execution Plane boundary — a logical distinction; physical deployment is governed by NCIE-004 where already established.

SOURCE	RELATIONSHIP	TARGET
Orchestration Control Plane	governs eligibility, policy, VPF requirements and routing for	Agent Execution Plane
Agent Execution Plane	executes the bounded runtime workload; never administers	Agent Registry, Model Registry, Tool Registry, VPF Policies or Agent Factory governance

24.4A.9 Architectural Flows

Figure 23. Existing agent path — ARGUS uses an eligible Registered Reusable Agent.

01	ARGUS
02	Agent Capability Discovery
03	Eligible Registered Agent
04	Model Gateway / Context / Tool Gateway
05	AI Result
06	VPF Gate B
07	ARGUS / Human

Figure 24. Dynamic synthesis path — ARGUS requests bounded Ephemeral Agent synthesis when no adequate registered capability exists.

01	ARGUS
02	No adequate eligible registered capability
03	Agent Factory
04	Agent Definition
05	VPF Gate A
06	Sandbox / Evaluation

	▼
07	Ephemeral Agent Runtime
	▼
08	Model Gateway / Context / Tool Gateway
	▼
09	Agent Result
	▼
10	VPF Gate B
	▼
11	ARGUS / Human

Figure 25. Capability-gap path — a missing primitive routes to engineering, never fabrication.

01	ARGUS
	▼
02	Required primitive unavailable
	▼
03	ENGINEERING CHANGE CANDIDATE
	▼
04	Human / Engineering / Codex process

24.4A.10 Gateway, Memory, Authority & Recursion Preservation

- AGENT → MODEL GATEWAY → ELIGIBLE MODEL, never AGENT → ARBITRARY PROVIDER.
- AGENT → TOOL GATEWAY → AUTHORIZED TOOL, never AGENT → DIRECT CONSEQUENTIAL SYSTEM ACCESS. Dynamic synthesis creates no bypass path for either gateway.
- Agents obtain governed context exclusively through the existing NCIE Memory/Context architecture (Ch.20); no unrestricted agent access to Institutional Memory, independent permanent agent memory, or agent ownership of NCIE Memory is depicted or specified. AGENT RUNTIME STATE ≠ INSTITUTIONAL MEMORY — detailed Memory semantics remain in NCIE-006.
- The effective capability of an agent remains bounded by the intersection of Current Human/Task Authorization, Agent Definition, Model Eligibility, Tool Policy, Data Classification, Context Policy and Runtime Policy; no component may expand authority independently.
- AGENT SYNTHESIS ≠ SELF-REGISTRATION: the Agent Factory may instantiate an Ephemeral Agent and ARGUS may recommend promotion, but the synthesized agent may not register itself — permanent reusable registration remains a governed lifecycle action defined in detail by NCIE-007 and governance documents.
- SYNTHESIZED AGENT ≠ AGENT-FACTORY AUTHORITY: Ephemeral Agents do not recursively create new agents by default. If recursive synthesis is ever approved, it requires a separate governed policy.

24.5 Model-Specific and Collaborative Evaluation (v0.4)

Before Participatory Mode (Ch.20 §20.7) is enabled in production for a given model/provider, the Collaborative Evaluation Harness must evaluate:

- Context tracking accuracy across a multi-turn, multi-speaker discussion.
- Multi-speaker attribution correctness (Ch.20 §20.6).
- Thread separation under concurrent analytical threads (Ch.20 §20.6).
- Contradiction-seeking and evidence-gap-identification quality (Ch.20 §20.3).
- Restraint — non-dominance under realistic conversational load.
- Human-control preservation — the model does not resist or work around a human narrowing or silencing its participation.

The Collaborative Quality Monitor continues to track over-participation and under-participation in production as two distinct, separately alertable failure modes, since they require opposite corrections.

24.6 Model Replacement

Replacing or upgrading a model requires it to reconnect to the governed Collaborative Reasoning State (Ch.20) and Evidence Service (Ch.23) through the same Model Gateway contract — a new model is never permitted to become an independent owner of collaborative or platform state.

24.7 Data Model

- Model Entry — provider, capability class, data-residency classification, training-use policy, approval status.
- Evaluation Result — model, evaluation suite, scores, pass/fail, approver.
- Guardrail Event — blocked or flagged request/response, rule triggered, timestamp.

24.8 Security Controls

- Every external model call is logged with the minimal context actually sent, for audit (Ch.23).
- A model/provider not present in the Model Registry with an approved status cannot be reached via the Model Gateway.
- Model Registry entries, evaluation results and approvals key off the Assistant's stable internal service identity, not its user-facing presentation name (ARGUS); renaming the presented Assistant does not invalidate or require reissuing prior approvals (Ch.19 §19.11).

24.9 Failure Modes

- Approved model/provider outage: Model Gateway fails over to a configured alternate approved model where available, or degrades to non-conversational tool access (Ch.19 §19.8, Ch.28).
- (v0.5) Agent Factory unavailable: existing eligible Registered Agents, direct governed ARGUS model use and deterministic NCIE remain available; new Dynamic Agent Synthesis returns an explicit synthesis-unavailable state (Ch.28 §28.3).
- (v0.5) VPF unavailable: substantive AI output requiring VPF remains UNVALIDATED or VALIDATION_UNAVAILABLE; VPF failure shall not fail open (§24.4A.4; Ch.28 §28.3).
- (v0.5) Model, Agent, Tool or VPF Profile Registry unavailable: eligibility is never assumed absent confirmed current Registry state; the platform fails safely or uses an explicitly approved current signed/cached state only where such a policy is already established (Ch.28 §28.3).

24.10 Non-Functional Requirements

- Guardrail evaluation adds no more than 150ms at p95 to a standard model request (Ch.30).

24.11 Acceptance Criteria

- A prompt-injection attempt embedded in ingested content is demonstrably treated as inert data, not as an instruction.
- A model/provider without a passing Collaborative Evaluation Harness result is demonstrably blocked from Participatory Mode production use.
- (v0.5) Architecture-level Dynamic Agent Synthesis and VPF acceptance scenarios are defined in Ch.31 §31.3A, including the mandatory Human-Primary adversarial scenario.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm internal-vs-external-vs-hybrid model deployment policy and data-residency constraints with NCA information-security and legal leadership; proposed default is internal/sovereign-hosted models preferred, with external models permitted only for non-sensitive rooms under the Context Minimizer and explicit Model Registry approval.

Traceability: NCIE-001 §NCIE Intelligence Assistant; §Security, Privacy, Identity, Authorization & Intelligence Sovereignty Governance; NCIE-001 v1.3 §12.80A, §18.241A; NCIE-007 v1.1 Ch.11, Ch.12, Ch.26.

API, Event Bus & Integration Contracts

Defines reliable inter-service communication.

Full production content. Traces to NCIE-001 v1.2 Data Acquisition, Integration, Source Intelligence & Ingestion Governance.

25.1 Scope

Defines reliable inter-service communication: governed APIs, the platform event bus, schema/contract versioning, and authorization enforcement across every service call — internal or external.

25.2 Principal Components

Table 29. API, Event Bus & Integration Contracts principal components.

Component	Responsibility
API Gateway	Single authenticated entry point for synchronous service and external-partner APIs.
Event Bus	Publishes PM/source/alert/workflow events for asynchronous consumption.
Schema Registry	Versions request/response and event schemas; enforces compatibility rules.
Service Contracts	Documented, versioned interface definitions between NCIE services.

25.3 Event Categories

- Source events — new artifact validated, source health change (Ch.6).
- PM/KPI events — degradation detected, investigation opened (Ch.7).
- Alert/Situation events — Alert raised/closed, Situation opened (Ch.18).
- Workflow events — Run completed, WAITING_FOR_* entered/resolved (Ch.21).
- Collaborative events — material collaboration audit events (Ch.20, Ch.23).

25.4 Reliability Patterns

- Every API call and event carries a correlation ID for end-to-end tracing.
- Event delivery is idempotent-consumer safe: duplicate delivery does not duplicate effect (Ch.21).
- Schema changes follow additive-first compatibility; breaking changes require a new major contract version, never an in-place redefinition.

25.5 Data Model

- Service Contract — name, version, request/response schema reference, owning service.
- Event — type, schema version, payload, correlation ID, published timestamp.

25.6 Security Controls

- The API Gateway enforces the same authorization layers as any other NCIE entry point (Ch.4); no internal service is reachable by bypassing it.
- External-partner API access uses scoped machine identities with the narrowest capability set required (Ch.4).

25.7 Failure Modes

- Event Bus degraded: publishing services buffer locally and retry; no event is silently dropped (Ch.28).

25.8 Non-Functional Requirements

- Internal API calls complete within 200ms at p95 under normal load; event delivery latency is bounded per event category (Ch.30).

25.9 Acceptance Criteria

- A schema-breaking change is demonstrably rejected unless published as a new contract version.
- A simulated duplicate event delivery demonstrably does not duplicate downstream effect.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm whether NCA has existing enterprise middleware/ESB standards that NCIE must integrate with rather than operate independently; the API Gateway/Event Bus architecture can front an existing middleware layer without redesign if required.

Traceability: NCIE-001 §Data Acquisition, Integration, Source Intelligence & Ingestion Governance.

Security, Privacy, Sovereignty & Sensitive Data

Defines technical protection of sensitive intelligence.

Full production content. Traces to NCIE-001 v1.2 Security, Privacy, Identity, Authorization & Intelligence Sovereignty Governance.

26.1 Scope

Defines the technical protection of sensitive intelligence beyond identity/authorization (Ch.4): encryption, secrets management, segmentation, protected-field handling, search/cache isolation, external-processing governance, and break-glass/export/redaction controls.

26.2 Principal Components

Table 30. Security, Privacy, Sovereignty & Sensitive Data principal components.

Component	Responsibility
Security Services	Platform-wide encryption, network segmentation and hardening baseline.
Secrets & Key Management	Centralized credential/key storage; no secret is ever embedded in code, config files or documents.
Data Protection Layer	Applies classification-aware handling (public/internal/protected/sensitive) across storage, search and cache.
Monitoring	Security event detection feeding Observability (Ch.27).

26.3 Classification and Protected Data

- Ghana Card and foreign-passport-derived identifiers (Ch.14), SIM identifiers, and precise location data are classified protected or sensitive and are field-level access-controlled (Ch.4).
- Search indexes, vector stores and caches are isolated per classification tier — a protected field is never indexed into a general-purpose search or embedding store without the same access control applied to the index itself.
- External AI/RPA processing of protected or sensitive data is governed by explicit policy in the Model Registry (Ch.24) and Capability Registry (Ch.21); default is minimisation and, where feasible, exclusion.

26.4 Break-Glass, Export and Redaction

- Break-glass access to normally restricted data is time-bounded, individually logged, and requires post-hoc justification review (Ch.4 §4.5).
- Data export is a controlled action requiring authorization and is itself an audited event (Ch.23).
- Redaction of protected fields for lower-clearance views is applied server-side, never left to client-side masking alone.
- Chain-of-custody for exported or redacted material is preserved via the Evidence/Provenance architecture (Ch.23).

26.5 Sovereignty Posture

Consistent with Chapter 1's posture statement, NCIE's default architecture keeps canonical and protected data within Ghanaian jurisdiction; any external processing is minimised, logged and policy-governed rather than assumed.

26.6 Data Model

- Classification Tag — field/record reference, tier (public/internal/protected/sensitive), applicable controls.
- Break-Glass Event — actor, accessed data, justification, review status.
- Export Record — requester, authorization reference, exported content reference, timestamp.

26.7 Security Controls

- Encryption at rest and in transit is mandatory for protected and sensitive tiers; keys are managed by the Secrets & Key Management component, never application code.
- Network segmentation isolates ingestion, canonical/analytical storage and external-facing services into distinct zones.

26.8 Failure Modes

- Key Management unavailable: services relying on protected-tier decryption fail closed rather than falling back to an unencrypted path (Ch.28).

26.9 Non-Functional Requirements

- Classification-tier lookups add no more than 20ms at p95 to a protected-field query (Ch.30).

26.10 Acceptance Criteria

- A protected field is demonstrably absent from any general-purpose search/embedding index accessible to unauthorized roles.
- A break-glass access is demonstrably logged with justification captured before or immediately after use.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm final data classification tiers, residency, retention and third-party processing policy with NCA data-protection/legal leadership; the architecture in this chapter is designed to be conservative pending that confirmation and does not require redesign once policy is finalised.

Traceability: NCIE-001 §Security, Privacy, Identity, Authorization & Intelligence Sovereignty Governance.

Observability, Health, Operations & Support

Defines how NCIE itself is monitored.

Full production content. Traces to NCIE-001 v1.2 Governance, Auditability, Explainability, Human Oversight, Operational Resilience & Product Acceptance.

27.1 Scope

Defines how NCIE monitors itself: platform health, automation health, source health and sector (domain) health kept as distinct, individually visible signals, plus runbooks and escalation.

27.2 Principal Components

Table 31. Observability, Health, Operations & Support principal components.

Component	Responsibility
Observability Platform	Collects metrics, logs and traces across all NCIE services.
Health Service	Aggregates platform-level health for the National Operating Picture (Ch.18).
Source Monitor	Tracks per-source freshness and coverage status (Ch.6).
Automation Monitor	Tracks Orchestrator/RPA provider health and Run success rates (Ch.21).

27.3 Health Signal Categories

- Platform health — service availability, latency, error rates.
- Automation health — provider availability, Run success/failure rates, queue depth.
- Source health — per-source freshness, coverage status (Green/Amber/Grey per Ch.6 §6.9), last successful acquisition.
- Sector/domain health — per-domain (Network, Traffic, Revenue, MoMo, SIM, Anti-Fraud, Regulatory) data currency and quality summary.

27.4 Explainable Operational State

Every health signal shown to a human is paired with the freshness/last-validated-output metadata that explains it, and the Assistant (Ch.19) can explain authoritative operational state in natural language grounded in these same signals — it does not have a separate, potentially inconsistent view of health.

27.5 Data Model

- Health Signal — category, scope (service/source/domain), status, last-updated, contributing metrics.
- Runbook — trigger condition, procedure reference, owning team.
- Escalation Policy — severity, notification chain, timeout-to-escalate.

27.6 Security Controls

- Observability data itself is classification-aware; logs/traces do not capture protected-tier field values in plaintext (Ch.26).

27.7 Failure Modes

- Observability Platform degraded: services continue operating; only visibility into their health is reduced, with that reduction itself surfaced as a meta-level health signal (Ch.28).

27.8 Non-Functional Requirements

- Health signal propagation from source event to visible status update completes within 60 seconds at p95 (Ch.30).

27.9 Acceptance Criteria

- Platform, automation, source and sector health are demonstrably shown as distinct signals for the same underlying incident, not collapsed into one status.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm the on-call/support model (in-house NCA team, vendor-supported, or hybrid) and required operational dashboards with NCA operations leadership; runbook content is produced during Ch.32 handover regardless of model chosen.

Traceability: NCIE-001 §Governance, Auditability, Explainability, Human Oversight, Operational Resilience & Product Acceptance.

Resilience, High Availability, Backup & Disaster Recovery

Defines failure behavior and recovery.

Full production content. Traces to NCIE-001 v1.2 Governance, Auditability, Explainability, Human Oversight, Operational Resilience & Product Acceptance.

28.1 Scope

Defines failure behaviour and recovery so that loss of any single external dependency degrades only the affected capability, never core NCIE, consistent with the graceful-degradation principle in Chapter 1.

28.2 Principal Components

Table 32. Resilience, High Availability, Backup & Disaster Recovery principal components.

Component	Responsibility
HA Design	Redundancy and failover design per service tier.
Backup	Scheduled, verified backups of canonical data, rules, audit, memory, schedules and workflows.
Disaster Recovery (DR)	Recovery procedures and secondary site/capability for platform-level failure.
Recovery Coordinator	Orchestrates ordered service restoration and state reconciliation after an outage.
State Reconciliation	Resolves state drift between primary and recovered/failed-over components.

28.3 Dependency Degradation Matrix

Table 33. What is preserved versus degraded when a given external dependency is lost.

Dependency Lost	Preserved Capability	Degraded Capability
AI / Model Gateway	Core NCIE, all canonical data, workflows, non-conversational tool access	Conversational Assistant, proactive room participation (Ch.19, Ch.20, Ch.24)
RPA provider	Manually triggered/API-based acquisition, all canonical data	Automated acquisition/execution via that provider; failover to alternate provider where configured (Ch.21)
Mapping provider	Full non-map workspace, all domain data and topology data	Map visualisation only; non-map geography selector remains available (Ch.3, Ch.8)
Single source system	All other sources, all prior canonical data	New data from that source; existing status visibly degraded, not silently missing (Ch.6)
(v0.5) Agent Factory	Existing eligible Registered Agents; direct governed ARGUS model use; deterministic NCIE	New Dynamic Agent Synthesis; an explicit synthesis-unavailable state is returned (Ch.24 §24.4A.1)
(v0.5) Ephemeral Agent Runtime	Existing unaffected services; Registered Agent execution	Dynamic Agent execution; an explicit runtime-unavailable state is returned, never a silent retry (Ch.24 §24.4A.3)
(v0.5) VPF	Deterministic NCIE functions; existing already-validated results	New substantive AI output remains UNVALIDATED or VALIDATION_UNAVAILABLE — VPF failure shall not fail open (Ch.24 §24.4A.4)
(v0.5) Model / Agent / Tool / VPF Profile Registry	Last-known eligible state only where an explicitly approved current signed/cached policy already exists	New eligibility determinations; no eligibility is assumed absent confirmed current Registry state (Ch.24 §24.9)

28.4 Backup Scope

- Canonical and analytical data (Ch.5).
- Evidence and provenance graph (Ch.23).
- Rules and their version history (Ch.22).
- Audit trail (Ch.23).
- Memory — private, shared room, and institutional (Ch.20).
- Schedules and in-flight/waiting workflow state (Ch.21) — a waiting workflow must survive a platform restart or failover without losing its wait state.

28.5 Recovery Flow

Figure 26. Disaster recovery and state-reconciliation flow.

01	Failure or DR event detected (Observability, Ch.27)
02	Recovery Coordinator initiates ordered restoration per service tier
03	Backup/replica promoted where required
04	State Reconciliation resolves drift against last-known-good canonical state
05	Waiting workflows resume from persisted wait state (Ch.21)
06	Health Service confirms restored status to National Operating Picture (Ch.18)

28.6 Security Controls

- Backups are encrypted and classification-aware, subject to the same protected-tier controls as live data (Ch.26).

28.7 Failure Modes

- Partial failover leaving two components with divergent state: State Reconciliation flags the divergence for review rather than auto-resolving silently.

28.8 Non-Functional Requirements

- Recovery Time Objective (RTO) and Recovery Point Objective (RPO) are defined per critical service tier (see proposed defaults below) and validated by DR test per Ch.31.

28.9 Acceptance Criteria

- A simulated loss of each dependency in §28.3 demonstrably preserves the stated capability and only degrades the stated one.
- A simulated platform restart during an active WAITING_FOR_PM workflow demonstrably resumes correctly rather than losing the wait state.
- (v0.5) A simulated VPF outage demonstrably leaves new substantive AI output UNVALIDATED or VALIDATION_UNAVAILABLE rather than PASS, while deterministic NCIE functions remain available (§28.3).

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Proposed default RTO/RPO: Tier 1 (identity, canonical data, evidence/audit) RTO 4 hours / RPO 15 minutes; Tier 2 (domain services, automation) RTO 8 hours / RPO 1 hour; Tier 3 (AI/collaborative features) RTO 24 hours / RPO 4 hours. Confirm against NCA business-continuity requirements and the DR site location.

Traceability: NCIE-001 §Governance, Auditability, Explainability, Human Oversight, Operational Resilience & Product Acceptance; NCIE-007 v1.1 Ch.32 (Model & Agent Lifecycle Governance).

Deployment, Environments, DevSecOps & Configuration

Defines controlled build and deployment.

Full production content. Traces to NCIE-001 v1.2 Governance, Auditability, Explainability, Human Oversight, Operational Resilience & Product Acceptance.

29.1 Scope

Defines controlled build and deployment: environment separation, versioned releases, secrets handling, rollback and the protection of sensitive production data from non-production environments.

29.2 Principal Components

Table 34. Deployment, Environments, DevSecOps & Configuration principal components.

Component	Responsibility
CI/CD	Automated build, test and deployment pipeline with mandatory gate checks (Ch.31).
Artifact Registry	Versioned, immutable build artifacts.
Environment Configuration	Per-environment configuration, separated from code and from production secrets.
Secrets Integration	Runtime secret injection from the Secrets & Key Management component (Ch.26); never committed to code or docs.

29.3 Environment Model

- Development — active engineering work; synthetic or heavily masked data only.
- Test — automated and manual test execution (Ch.31); synthetic or masked data only.
- Staging — production-representative rehearsal environment; masked production-shaped data only.
- Production — live NCIE; the only environment permitted to hold real protected/sensitive data.

29.4 Release Discipline

- Every deployment is versioned and traceable to a specific artifact and commit.
- Rollback to the immediately prior artifact version is a supported, tested operation, not an emergency improvisation.
- Every deployment is itself an audited change event (Ch.23), recording who approved and executed it.

29.5 Data Model

- Release — artifact version, environment, deployed-by, timestamp, rollback target.
- Environment Config — environment, key, value reference (secrets stored by reference, not value).

29.6 Security Controls

- Secrets are never present in source control, CI/CD logs or any document in this suite; they are injected at runtime by reference only.
- Production data does not flow to non-production environments unmasked; masking is enforced by the pipeline, not left to manual discipline.

29.7 Failure Modes

- Failed deployment gate check: deployment is blocked from promotion; the pipeline does not allow manual override of a failed mandatory gate without a recorded exception (Ch.31).

29.8 Non-Functional Requirements

- Rollback to the prior artifact version completes within the RTO defined for the affected service tier (Ch.28).

29.9 Acceptance Criteria

- A deployment with a failing mandatory gate is demonstrably blocked from reaching production.
- A rollback is demonstrably exercised successfully in a rehearsal environment before go-live.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm the hosting model (on-premises NCA data centre, sovereign cloud, or hybrid — consistent with the posture proposed in Ch.1 §1.6) and release authority (who may approve a production deployment) with NCA IT leadership.

Traceability: NCIE-001 §Governance, Auditability, Explainability, Human Oversight, Operational Resilience & Product Acceptance.

Performance, Scalability, Accessibility & Non-Functional Architecture

Defines measurable quality attributes.

Full production content. Traces to NCIE-001 v1.2 Product Capability Architecture; Governance, Auditability, Explainability, Human Oversight, Operational Resilience & Product Acceptance.

30.1 Scope

Defines the measurable quality attributes referenced by number throughout this specification (e.g. "p95 latency", "national data volumes") as one consolidated, consistent set.

30.2 Reference Non-Functional Targets

Table 35. Reference non-functional targets consumed by other chapters (proposed baseline; confirm at scale).

Attribute	Target	Applies To
Interactive query latency (p95)	≤ 3 seconds	Workspace panels, Assistant invoked responses, Fusion queries (Ch.3, Ch.17, Ch.19)
Authorization decision latency (p95)	≤ 100 ms	Authorization Service (Ch.4)
Internal API latency (p95)	≤ 200 ms	API Gateway (Ch.25)
Scheduled Request punctuality (p99)	≤ 1 minute drift	Scheduler (Ch.21)
Health signal propagation (p95)	≤ 60 seconds	Observability (Ch.27)
Concurrent users (initial production target)	≥ 500 concurrent, ≥ 3,000 registered	Platform-wide
National cell/topology scale	≥ current national cell count, versioned	Ch.7, Ch.8
Accessibility conformance	WCAG 2.1 AA	Situational Awareness Workspace (Ch.3)

30.3 Capacity Model

Canonical and analytical stores scale horizontally by operator/geography/time partition (Ch.5); concurrency controls at the Authorization and API layers (Ch.4, Ch.25) prevent a single tenant or workload from starving others. Capacity planning is revisited at each major release using Observability trend data (Ch.27).

30.4 Accessibility Beyond Colour

- Every status distinction that uses colour (e.g. Green/Amber/Grey coverage, Ch.6 §6.9) also carries a text label and, where applicable, an icon shape — colour is never the only signal.
- Keyboard navigation and screen-reader labelling are required for all primary Situational Awareness Workspace panels (Ch.3).

30.5 Performance Cannot Bypass Governance

No performance optimisation (caching, precomputation, denormalisation) is permitted to bypass authorization filtering, evidence/provenance recording, or audit logging. Where caching is used for latency (e.g. Ch.4 authorization decisions, Ch.22 rule lookups), cache invalidation is itself designed to preserve correctness of these controls, not just data freshness.

30.6 Security Controls

- Load-testing and capacity-testing environments use masked data only, consistent with Ch.29 §29.3.

30.7 Failure Modes

- Load exceeding capacity targets: services degrade gracefully with request shedding/backpressure and visible status, rather than uncontrolled failure (Ch.28).

30.8 Acceptance Criteria

- Each target in §30.2 is demonstrably measured under a representative load test and recorded as part of production acceptance (Ch.31).
- A representative workspace panel passes an automated WCAG 2.1 AA accessibility check.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. The specific figures in §30.2 (concurrent users, national cell count, latency targets) are proposed baselines derived from NCIE-001's stated scale ambitions; confirm exact production-scale targets with NCA capacity-planning stakeholders before final acceptance sign-off.

Traceability: NCIE-001 §Product Capability Architecture; §Governance, Auditability, Explainability, Human Oversight, Operational Resilience & Product Acceptance.

Testing, Verification, Traceability & Acceptance Architecture

Defines how every requirement will be proven.

Full production content. Traces to NCIE-001 v1.2 Governance, Auditability, Explainability, Human Oversight, Operational Resilience & Product Acceptance.

31.1 Scope

Defines how every requirement in NCIE-001 and every architectural commitment in NCIE-002 will be proven, through a Requirements Traceability Matrix (RTM), layered test suites, and governed acceptance evidence.

31.2 Principal Components

Table 36. Testing, Verification, Traceability & Acceptance Architecture principal components.

Component	Responsibility
Test Architecture	Defines test layers, environments and required coverage per chapter.
Requirements Traceability Matrix (RTM)	Maps every NCIE-001 requirement ID (chapter.paragraph) to its NCIE-002 architectural component(s) and test case(s).
Evaluation Suites	AI- and automation-specific evaluation suites, including the Collaborative Evaluation Harness (Ch.24 §24.5).
Acceptance Evidence Repository	Stores test results, sign-offs and defect references as durable acceptance evidence.

31.3 Test Layers

- Functional — per-chapter feature correctness (Ch.6-Ch.22 domain behaviours).
- Integration — cross-service and cross-domain flows (Ch.17, Ch.25).
- Security — authorization, protected-data and guardrail verification (Ch.4, Ch.26, Ch.24).
- AI/Collaborative evaluation — Model Gateway and collaborative-participation quality (Ch.20, Ch.24).
- Automation — RPA/orchestration reliability, idempotency and failover (Ch.21).
- Resilience — dependency-loss and DR rehearsal (Ch.28).
- Cross-domain and multi-user — scenarios spanning multiple domains and multiple concurrent, differently-permissioned users (Ch.17, Ch.20).

31.3A Dynamic Agent Synthesis & VPF Acceptance Architecture (v0.5)

Architecture-level acceptance scenarios for the Ch.24 §24.4A extension. These are testable architectural acceptance criteria, not the full NCIE-007 acceptance catalogue (NCIE-007 v1.1 Ch.32 remains authoritative for the detailed test suite):

- ARGUS can use an eligible Registered Agent.
- ARGUS can synthesize a bounded Ephemeral Agent when no adequate registered agent exists.
- Individual bounded synthesis does not require Codex hand-coding.
- A synthesized agent cannot create new permissions.
- A synthesized agent cannot self-register.
- Recursive synthesis is disabled by default.
- A synthesized agent cannot bypass the Model Gateway.
- A synthesized agent cannot bypass the Tool Gateway.
- Substantive AI output cannot bypass VPF.
- VPF PASS ≠ Human / Institutional Approval.
- VPF unavailable ≠ PASS.
- Agent Factory unavailable does not disable deterministic NCIE.
- A missing primitive produces an Engineering Change Candidate.
- Current authorization remains authoritative at execution.

31.3B Human-Primary Acceptance Scenario (Mandatory)

One mandatory adversarial acceptance scenario, which must pass before production acceptance: ARGUS receives a high-consequence task; ARGUS synthesizes an agent; VPF Gate A passes; the sandbox passes; the agent uses eligible models and tools; VPF Gate B returns PASS; several agents agree. Verify that the consequential institutional action still requires the applicable current human/institutional authorization — none of the preceding facts independently creates institutional execution authority (Ch.24 §24.4A.10).

31.4 Requirements Traceability Matrix

The RTM uses NCIE-001's stable chapter.paragraph requirement ID (e.g. §18.241, per NCIE-001 §18.241 "Requirement ID") as the anchor, and maps each to the NCIE-002 chapter/component that implements it and the test case(s) that verify it. This chapter defines the RTM's structure; the populated RTM is a companion artifact maintained alongside this document and delivered as part of Chapter 32 handover.

Table 37. Requirements Traceability Matrix structure.

RTM Column	Content
Requirement ID	NCIE-001 chapter.paragraph reference (e.g. §18.241).
Requirement Summary	Short restatement of the requirement.
NCIE-002 Component	Owning architectural component (e.g. Rule Service, Ch.22).
Test Case ID	Reference to the verifying test case.
Status	Not started / In progress / Passed / Failed / Waived.
Evidence Reference	Link into the Acceptance Evidence Repository.

31.5 Acceptance Gates

- A critical defect (data corruption, authorization bypass, evidence/provenance loss, human-authority bypass) blocks ordinary production acceptance — no severity-based waiver is permitted for these categories.
- AI shall not approve NCIE for production (per NCIE-001 §18.244); final acceptance authority is always a named human/institutional authority.
- Conditional acceptance, where permitted for non-critical gaps, records Known Limitation, Risk, Mitigation, Owner and Target Resolution (per NCIE-001 §18.243).

31.6 Data Model

- RTM Entry — as per §31.4 table.
- Test Case — id, layer, preconditions, steps, expected result, linked requirement ID(s).
- Acceptance Record — test case, executed-by, date, result, defect reference if applicable.

31.7 Security Controls

- Test environments never contain unmasked production protected/sensitive data (Ch.29 §29.3).

31.8 Failure Modes

- A failed mandatory acceptance criterion is not marked passed without corrective evidence attached (per NCIE-001 §18.242).

31.9 Acceptance Criteria for This Chapter

- The RTM structure in §31.4 is demonstrably populated for at least one representative requirement per NCIE-002 chapter as a proof of traceability before full production acceptance.
- Five-domain acceptance (Traffic, Revenue, Mobile Money, SIM Registration, Anti-Fraud, per NCIE-001 §18.245) is explicitly represented in the RTM.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm acceptance authorities per domain (who signs off Five-Domain Acceptance, who signs off overall production acceptance) with NCA leadership; NCIE-001 already fixes that AI may not hold this authority.

Traceability: NCIE-001 §Governance, Auditability, Explainability, Human Oversight, Operational Resilience & Product Acceptance; NCIE-007 v1.1 Ch.32 (Model & Agent Lifecycle Governance) §32.7-§32.10.

Operations, Administration, Documentation & Codex Handover

Defines sustainable production ownership and the package handed to Codex/implementation teams.

Full production content. Traces to NCIE-001 v1.2 Governance, Auditability, Explainability, Human Oversight, Operational Resilience & Product Acceptance.

32.1 Scope

Defines sustainable production ownership and the package handed to Codex/implementation teams: administration surfaces, runbooks, the full documentation set, and an implementation backlog mapped to requirements.

32.2 Principal Components

Table 38. Operations, Administration, Documentation & Codex Handover principal components.

Component	Responsibility
Administration Plane	Admin surfaces for users/roles, rules, sources, automation providers and AI models.
Documentation Set	Architecture (this document), API, data and security specifications (NCIE-003 through NCIE-016).
Runbooks	Operational procedures per Ch.27/Ch.28, with named owners.
Handover Package	Consolidated deliverable to Codex: architecture, RTM, deployment/recovery docs, backlog — no credentials included.
Implementation Work Breakdown	NCIE-002 components mapped to build sequence (detailed in NCIE-017).

32.3 Administration Surfaces

- User and role administration (Ch.4).
- Rule administration (Ch.22): surcharge rate, SIM limits, KPI thresholds, REWS rules.
- Source administration (Ch.6): registry entries, owners, health thresholds.
- Automation administration (Ch.21): Capability Registry entries, provider policy.
- Model administration (Ch.24): Model Registry entries, evaluation status, participation-mode approval per model.

32.4 Handover Contents

- This document (NCIE-002) as the authoritative architecture reference.
- The populated Requirements Traceability Matrix (Ch.31).
- Deployment and recovery documentation (Ch.28, Ch.29).
- Runbooks with named operational owners (Ch.27).
- Implementation backlog mapped to NCIE-001 requirement IDs and NCIE-002 components.
- No credentials, secrets or keys are included in any handover artifact — these are provisioned separately through the Secrets & Key Management component (Ch.26).

32.4A Dynamic Agent Synthesis & VPF Engineering Handover (v0.5)

Extends the implementation handover to explicitly require engineering delivery of:

- Agent Factory (Ch.24 §24.4A.1).
- Agent Definition logical implementation.
- Ephemeral Agent Runtime (Ch.24 §24.4A.3).
- Agent Registry integration (Ch.24 §24.4A.2).
- VPF runtime (Ch.24 §24.4A.4).
- VPF Profile Registry integration (Ch.24 §24.4A.7).
- VPF Gate A (Ch.24 §24.4A.5).

- VPF Gate B (Ch.24 §24.4A.6).
- Sandbox/Evaluation capability.
- Runtime-policy enforcement.
- Agent expiry.
- Kill/Suspend controls.
- Provenance.
- Observability.

Final vendor choices are not prescribed here beyond what is already approved elsewhere (NCIE-004). Once these capabilities are implemented and operationally approved, ARGUS can create bounded task-specific agents without requiring bespoke Codex development for every individual agent. Where the required capability exceeds the approved Agent Factory primitives, engineering/Codex intervention is required (Ch.24 §24.4A.1).

32.5 Handover Flow

Figure 27. Architecture-to-Codex handover flow.

01	Architecture and RTM finalised and reviewed (Ch.1-31)
02	Administration Plane configured for initial production users/rules/sources
03	Runbooks assigned to named operational owners
04	Implementation Work Breakdown sequenced (NCIE-017)
05	Handover Package assembled, credentials excluded
06	Codex/implementation teams receive package and begin build

32.6 Data Model

- Admin Action — actor, target (user/rule/source/provider/model), change, timestamp (audited per Ch.23).
- Runbook Entry — trigger, procedure, owner, last-reviewed date.
- Handover Artifact — type, version, inclusion/exclusion status (e.g. credentials explicitly excluded).

32.7 Security Controls

- Administration Plane actions are themselves authorized and audited exactly like any other sensitive action (Ch.4, Ch.23) — administrator status is not a bypass of governance.
- Handover Package assembly is checked against a no-credentials checklist before release.

32.8 Failure Modes

- Runbook without a named current owner: flagged by the Administration Plane as an operational gap rather than silently left unassigned.

32.9 Non-Functional Requirements

- Administration Plane actions apply and take effect within the same latency targets as their underlying component (Ch.30).

32.10 Acceptance Criteria

- The Handover Package is demonstrably assembled with all items in §32.4 present and the no-credentials check passed.
- Every runbook has a named current owner at the time of handover.
- (v0.5) The §32.4A engineering deliverable list is demonstrably present in the Handover Package before Dynamic Agent Synthesis is activated in production.

⚠ PROPOSED — PENDING INSTITUTIONAL CONFIRMATION. Confirm required handover artifacts beyond the baseline in §32.4, and the operational owners for each runbook category, with NCA operations and the Codex/implementation team lead.

Traceability: NCIE-001 §Governance, Auditability, Explainability, Human Oversight, Operational Resilience & Product Acceptance; NCIE-001 v1.3 §18.241A; NCIE-007 v1.1 Ch.36 §36.7-§36.8.